

Solutions to
An Introduction to Quantum Field Theory
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Chapter 1

Invitation: Pair Production in e^+e^- Annihilation

No Problems

Chapter 2

The Klein-Gordon Field

2.1 Classical electromagnetism with no sources

The Euler-Lagrange equations of motion, with the components $A_\mu(x)$ as the dynamical variables, are given by:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0,$$

where the Lagrangian density is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu},$$

and

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

is the electromagnetic field tensor. For the Euler-Lagrange equations of motion, μ and ν are *free* indices that simply label the components A_μ and its derivatives $\partial_\mu A_\nu$. In contrast, the indices for $F_{\mu\nu}$ are *dummy* indices that are used to sum over all the components. Their label doesn't matter as long as it is distinct from the *free* indices. It is important to note that $F_{\mu\nu}$ is antisymmetric. More precisely,

$$F_{\nu\mu} = \partial_\nu A_\mu - \partial_\mu A_\nu = -(\partial_\mu A_\nu - \partial_\nu A_\mu) = -F_{\mu\nu}.$$

Consequently, the diagonal components of $F_{\mu\nu}$ vanish. Specifically,

$$F_{\mu\mu} = -F_{\mu\mu} \implies 2F_{\mu\mu} = 0 \implies F_{\mu\mu} = 0.$$

(a) For the second term in the Euler-Lagrange equations of motion, we have:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial A_\nu} &= \frac{\partial}{\partial A_\nu} \left(-\frac{1}{4} F_{\alpha\beta} F^{\alpha\beta} \right) \rightarrow \text{apply product rule} \\
&= -\frac{1}{4} \left(F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial A_\nu} + F_{\alpha\beta} \frac{\partial F^{\alpha\beta}}{\partial A_\nu} \right) \rightarrow \text{lower indices in 2nd term} \\
&= -\frac{1}{4} \left(F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial A_\nu} + g^{\alpha\sigma} g^{\beta\rho} F_{\alpha\beta} \frac{\partial F_{\sigma\rho}}{\partial A_\nu} \right) \rightarrow \text{raise indices in 2nd term} \\
&= -\frac{1}{4} \left(F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial A_\nu} + F^{\sigma\rho} \frac{\partial F_{\sigma\rho}}{\partial A_\nu} \right) \rightarrow \text{relabel } \sigma \text{ and } \rho \text{ to } \alpha \text{ and } \beta \\
&= -\frac{1}{4} \left(F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial A_\nu} + F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial A_\nu} \right) \rightarrow \text{terms are equal} \\
&= -\frac{1}{2} F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial A_\nu} \rightarrow \text{second term is 0} \\
&= 0.
\end{aligned}$$

This is because

$$\frac{\partial F_{\alpha\beta}}{\partial A_\nu} = \frac{\partial}{\partial A_\nu} (\partial_\alpha A_\beta - \partial_\beta A_\alpha) = 0,$$

which follows from the fact that $F_{\mu\nu}$ depends only on the derivative of A_μ (i.e., $\partial_\mu A_\nu$) and not A_μ itself. For the first term in the Euler-Lagrange equations of motion, we have:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} &= \frac{\partial}{\partial(\partial_\mu A_\nu)} \left(-\frac{1}{4} F_{\alpha\beta} F^{\alpha\beta} \right) \rightarrow \text{apply product rule} \\
&= -\frac{1}{4} \left(F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} + F_{\alpha\beta} \frac{\partial F^{\alpha\beta}}{\partial(\partial_\mu A_\nu)} \right) \rightarrow \text{lower indices in 2nd term} \\
&= -\frac{1}{4} \left(F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} + g^{\alpha\sigma} g^{\beta\rho} F_{\alpha\beta} \frac{\partial F_{\sigma\rho}}{\partial(\partial_\mu A_\nu)} \right) \rightarrow \text{raise indices in 2nd term} \\
&= -\frac{1}{4} \left(F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} + F^{\sigma\rho} \frac{\partial F_{\sigma\rho}}{\partial(\partial_\mu A_\nu)} \right) \rightarrow \text{relabel } \sigma \text{ and } \rho \text{ to } \alpha \text{ and } \beta \\
&= -\frac{1}{4} \left(F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} + F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} \right) \rightarrow \text{terms are equal} \\
&= -\frac{1}{2} F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} \rightarrow \text{insert } F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \\
&= -\frac{1}{2} F^{\alpha\beta} \frac{\partial}{\partial(\partial_\mu A_\nu)} (\partial_\alpha A_\beta - \partial_\beta A_\alpha) \rightarrow \mu, \nu = \alpha, \beta \text{ or } \beta, \alpha \text{ survive}
\end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2}F^{\alpha\beta}\left(\delta_{\mu\alpha}\delta_{\nu\beta}-\delta_{\mu\beta}\delta_{\nu\alpha}\right) \rightarrow \text{distribute } F^{\alpha\beta} \\
&= -\frac{1}{2}\left(F^{\alpha\beta}\delta_{\mu\alpha}\delta_{\nu\beta}-F^{\alpha\beta}\delta_{\mu\beta}\delta_{\nu\alpha}\right) \rightarrow \text{satisfy } \delta_{ij} \text{ functions} \\
&= -\frac{1}{2}\left(F^{\mu\nu}-F^{\nu\mu}\right) \rightarrow F_{\mu\nu} \text{ is antisymmetric} \\
&= -\frac{1}{2}\left(F^{\mu\nu}+F^{\mu\nu}\right) \\
&= -F^{\mu\nu}.
\end{aligned}$$

Inserting these results into the Euler-Lagrange equations of motion, we obtain:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0 \implies \partial_\mu (-F^{\mu\nu}) - 0 = 0 \implies \partial_\mu F^{\mu\nu} = 0.$$

Two of Maxwell's equations can be written in standard form by making use of this result. First, consider the identity $E^i = -F^{0i} = F^{i0}$, where $\nu = 0$:

$$\begin{aligned}
0 &= \partial_\mu F^{\mu 0} \rightarrow \text{separate components} \\
&= \partial_0 F^{00} + \partial_i F^{i0} \rightarrow F^{00} = 0 \text{ and substitute } F^{i0} = E^i \\
&= \partial_i E^i \rightarrow \text{divergence of a vector field in tensor notation} \\
&= \nabla \cdot \mathbf{E} \rightarrow \text{Gauss's law.}
\end{aligned}$$

Second, consider the identity $\epsilon^{ijk}B^k = -F^{ij} = F^{ji}$, where $\nu = i$:

$$\begin{aligned}
0 &= \partial_\mu F^{\mu i} \rightarrow \text{separate components} \\
&= \partial_0 F^{0i} + \partial_j F^{ji} \rightarrow \text{substitute } E^i \text{ and } B^k \\
&= -\partial_0 E^i + \epsilon^{ijk}\partial_j B^k \rightarrow \text{2nd term is curl of a vector field in tensor notation} \\
&= \frac{\partial \mathbf{E}}{\partial t} - \nabla \times \mathbf{B} \rightarrow \text{Ampere's law.}
\end{aligned}$$

The two remaining Maxwell equations can be written in standard form by making use of these identities with the definition of $F_{\mu\nu}$. First, consider the case with $\mu = i$ and $\nu = j$, which gives:

$$\begin{aligned}
F^{ij} &\implies \partial^i A^j - \partial^j A^i = -\epsilon^{ijk}B^k \rightarrow \text{multiply by } \epsilon^{ijm} \text{ and sum over } i, j \\
&-\epsilon^{ijm}(\partial_i A^j - \partial_j A^i) = -\epsilon^{ijm}\epsilon^{ijk}B^k \rightarrow \epsilon^{ijm}\epsilon^{ijk} = 2\delta^{mk} \\
&\epsilon^{ijm}\partial_i A^j - \epsilon^{ijm}\partial_j A^i = 2\delta^{mk}B^k \rightarrow \text{relabel } i, j \text{ to } j, i \text{ in LHS 2nd term} \\
&\epsilon^{ijm}\partial_i A^j - \epsilon^{jim}\partial_i A^j = 2\delta^{mk}B^k \rightarrow \text{swap } i, j \text{ in } \epsilon^{jim} \\
&\epsilon^{ijm}\partial_i A^j - (-\epsilon^{ijm}\partial_i A^j) = 2\delta^{mk}B^k
\end{aligned}$$

$$2\epsilon^{ijm}\partial_i A^j = 2\delta^{mk} B^k$$

$$\epsilon^{ijm}\partial_i A^j = B^m \rightarrow \text{curl of a vector field in tensor notation}$$

$$\nabla \times \mathbf{A} = \mathbf{B} \rightarrow \text{take the divergence and recall } \nabla \cdot (\nabla \times \mathbf{A}) = 0$$

$$0 = \nabla \cdot \mathbf{B} \rightarrow \text{Gauss's law for magnetism.}$$

Finally, consider the case with $\mu = 0$ and $\nu = i$, which gives:

$$F^{0i} \implies \partial^0 A^i - \partial^i A^0 = -E^i \rightarrow \text{lower an index in LHS 2nd term}$$

$$\partial^0 A^i + \partial_i A^0 = -E^i$$

$$\frac{\partial A^i}{\partial t} + \frac{\partial A^0}{\partial x^i} = -E^i \rightarrow \text{second term on LHS is the scalar potential}$$

$$-\frac{\partial \mathbf{A}}{\partial t} - \nabla \Phi = \mathbf{E} \rightarrow \text{take the curl}$$

$$-\frac{\partial}{\partial t} (\nabla \times \mathbf{A}) - \nabla \times (\nabla \Phi) = \nabla \times \mathbf{E} \rightarrow \nabla \times \mathbf{A} = \mathbf{B} \text{ from above}$$

$$-\frac{\partial \mathbf{B}}{\partial t} = \nabla \times \mathbf{E} \rightarrow \text{Faraday's law.}$$

(b) From Noether's theorem applied to spacetime transformations, where the components $A_\mu(x)$ are the dynamical variables, we have the general form of the energy-momentum tensor expressed as:

$$T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\lambda)} \partial_\nu A_\lambda - \mathcal{L} \delta^\mu_\nu.$$

Using information from part (a), this becomes:

$$T^\mu_\nu = -F^{\mu\lambda} \partial_\nu A_\lambda + \frac{1}{4} F_{\alpha\beta} F^{\alpha\beta} \delta^\mu_\nu \rightarrow \text{apply } g^{\sigma\nu}$$

$$g^{\sigma\nu} T^\mu_\nu = -F^{\mu\lambda} g^{\sigma\nu} \partial_\nu A_\lambda + \frac{1}{4} F_{\alpha\beta} F^{\alpha\beta} g^{\sigma\nu} \delta^\mu_\nu$$

$$T^{\mu\sigma} = -F^{\mu\lambda} \partial^\sigma A_\lambda + \frac{1}{4} g^{\sigma\mu} F_{\alpha\beta} F^{\alpha\beta} \rightarrow \text{relabel } \sigma \text{ to } \nu \text{ and use } g^{\mu\nu} = g^{\nu\mu}$$

$$T^{\mu\nu} = -F^{\mu\lambda} \partial^\nu A_\lambda + \frac{1}{4} g^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}.$$

We're told that this expression is not a symmetric tensor, but that we can remedy this by adding the term $\partial_\lambda K^{\lambda\mu\nu}$ to $T^{\mu\nu}$, where $K^{\lambda\mu\nu}$ is antisymmetric in the first two indices. The resulting expression is:

$$\begin{aligned} \hat{T}^{\mu\nu} &= T^{\mu\nu} + \partial_\lambda K^{\lambda\mu\nu} \\ &= -F^{\mu\lambda} \partial^\nu A_\lambda + \frac{1}{4} g^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} + \partial_\lambda K^{\lambda\mu\nu} \end{aligned}$$

$$= -F^{\mu\lambda}\partial^\nu A_\lambda + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} + \partial_\lambda F^{\mu\lambda}A^\nu,$$

where we used the provided relationship, $K^{\lambda\mu\nu} = F^{\mu\lambda}A^\nu$, in the last term. This expression can be further simplified as follows:

$$\begin{aligned}\widehat{T}^{\mu\nu} &= -F^{\mu\lambda}\partial^\nu A_\lambda + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} + \partial_\lambda F^{\mu\lambda}A^\nu \rightarrow \text{apply product rule} \\ &= -F^{\mu\lambda}\partial^\nu A_\lambda + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} + F^{\mu\lambda}\partial_\lambda A^\nu + A^\nu\partial_\lambda F^{\mu\lambda} \rightarrow \partial_\lambda F^{\mu\lambda} = 0 \\ &= -F^{\mu\lambda}\partial^\nu A_\lambda + F^{\mu\lambda}\partial_\lambda A^\nu + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} \\ &= -F^{\mu\lambda}(\partial^\nu A_\lambda - \partial_\lambda A^\nu) + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} \\ &= -F^{\mu\lambda}F^\nu{}_\lambda + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}.\end{aligned}$$

The symmetry of this expression under the exchange $\mu \leftrightarrow \nu$ can be demonstrated as follows:

$$\begin{aligned}\widehat{T}^{\nu\mu} &= -F^{\nu\lambda}F^\mu{}_\lambda + \frac{1}{4}g^{\nu\mu}F_{\alpha\beta}F^{\alpha\beta} \rightarrow \text{raise and lower indices and use } g^{\mu\nu} = g^{\nu\mu} \\ &= -g^{\lambda\sigma}g_{\lambda\rho}F^\nu{}_\sigma F^{\mu\rho} + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} \rightarrow g^{\lambda\sigma}g_{\lambda\rho} = \delta^\sigma{}_\rho \\ &= -F^\nu{}_\sigma F^{\mu\sigma} + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} \rightarrow \text{relabel } \sigma \text{ to } \lambda \\ &= -F^\nu{}_\lambda F^{\mu\lambda} + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} \\ &= \widehat{T}^{\mu\nu}.\end{aligned}$$

The electromagnetic energy density, ε , is obtained from $\widehat{T}^{00}$:

$$\begin{aligned}\varepsilon &= \widehat{T}^{00} \\ &= -F^{0\lambda}F^0{}_\lambda + \frac{1}{4}g^{00}F_{\alpha\beta}F^{\alpha\beta} \rightarrow \text{separate components in 1st term} \\ &= -F^{00}F^0{}_0 - F^{0i}F^0{}_i + \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta} \rightarrow F^{00} = F^0{}_0 = 0 \text{ and } E^i = -F^{0i} = F^0{}_i \\ &= E^i E^i + \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta} \rightarrow \text{separate components in 2nd term} \\ &= E^i E^i + \frac{1}{4}\left(F_{00}F^{00} + F_{0i}F^{0i} + F_{i0}F^{i0} + F_{ij}F^{ij}\right) \rightarrow F_{00} = F^{00} = 0 \\ &= E^i E^i + \frac{1}{4}\left(E^i(-E^i) + (-E^i)E^i + (-\epsilon^{ijk}B^k)(-\epsilon^{ijl}B^l)\right) \\ &= E^i E^i + \frac{1}{4}\left(-2E^i E^i + \epsilon^{ijk}\epsilon^{ijl}B^k B^l\right) \rightarrow \epsilon^{ijk}\epsilon^{ijl} = 2\delta^{kl}\end{aligned}$$

$$\begin{aligned}
&= E^i E^i + \frac{1}{4} \left(-2E^i E^i + 2B^k B^k \right) \\
&= \frac{1}{2} E^i E^i + \frac{1}{2} B^k B^k \\
&= \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2).
\end{aligned}$$

The momentum density, $\mathbf{S}$, is obtained from $\hat{T}^{0i}$:

$$\begin{aligned}
\mathbf{S} &= \hat{T}^{0i} \\
&= -F^{0\lambda} F^i_{\lambda} + \frac{1}{4} g^{0i} F_{\alpha\beta} F^{\alpha\beta} \rightarrow g^{0i} = 0 \\
&= -F^{00} F^i_0 - F^{0j} F^i_j \rightarrow F^{00} = 0 \\
&= (E^j) (\epsilon^{ijk} B^k) \\
&= \epsilon^{ijk} E^j B^k \\
&= \mathbf{E} \times \mathbf{B}.
\end{aligned}$$

2.2 The complex scalar field

The action for a field theory of a complex-valued scalar field obeying the Klein-Gordon equation is given by:

$$S = \int d^4x \left(\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi \right).$$

The authors suggest it is easiest to consider the fields $\phi(x)$ and $\phi^*(x)$ as the basic dynamical variables rather than decomposing $\phi(x)$ into its real and imaginary parts, $\phi(x) = \phi_1(x) + i\phi_2(x)$, where $\phi_1(x)$ and $\phi_2(x)$ are real Klein-Gordon fields. Therefore, we need an appropriate expression for the Fourier mode expansion of $\phi(x)$ to describe complex-valued fields. The position-space representation of a general scalar field $\phi(x)$ is given by its inverse Fourier transform:

$$\phi(x) = \int \frac{d^4p}{(2\pi)^4} \tilde{\phi}(p) e^{-ip \cdot x},$$

where $\tilde{\phi}(p)$ is the Fourier transform of the field in momentum space. Inserting this expression into the Klein-Gordon equation, we obtain:

$$\begin{aligned}
0 &= \left(\partial_\mu \partial^\mu + m^2 \right) \phi(x) \\
&= \int \frac{d^4p}{(2\pi)^4} \left(m^2 - p^2 \right) \tilde{\phi}(p) e^{-ip \cdot x}.
\end{aligned}$$

This expression implies the *on-mass-shell* condition for $\tilde{\phi}(p)$. More precisely, in order to satisfy the Klein-Gordon equation, $\tilde{\phi}(p)$ must vanish everywhere except where the four-momentum

satisfies $p^2 = m^2$. Consequently, the spectrum takes the general form $\tilde{\phi}(p) = 2\pi\delta(p^2 - m^2)f(p)$, for some general function $f(p)$. The factor of 2π ensures consistency with normalization conventions. Given that $p^2 = (p^0)^2 - \mathbf{p}^2$, the argument of the Dirac delta function can be rewritten as $\delta((p^0)^2 - (E_{\mathbf{p}})^2)$, where $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$. As a result, the Dirac delta function has two roots at $p^0 = \pm E_{\mathbf{p}}$, so we can apply the identity:

$$\delta(f(x)) = \sum_i \frac{1}{|f'(x_i)|} \delta(x - x_i)$$

to obtain

$$\tilde{\phi}(p) = \frac{2\pi}{2E_{\mathbf{p}}} \left(\delta(p^0 - E_{\mathbf{p}}) + \delta(p^0 + E_{\mathbf{p}}) \right) f(p).$$

Inserting this expression for $\tilde{\phi}(p)$ into the above expression for $\phi(x)$, and evaluating the p^0 integral, we obtain:

$$\begin{aligned} \phi(x) &= \int \frac{d^4p}{(2\pi)^4} \frac{2\pi}{2E_{\mathbf{p}}} \left(\delta(p^0 - E_{\mathbf{p}}) + \delta(p^0 + E_{\mathbf{p}}) \right) f(p) e^{-ip \cdot x} \\ &= \int \frac{d^3p}{(2\pi)^3} \int \frac{dp^0}{2\pi} \frac{2\pi}{2E_{\mathbf{p}}} \left(\delta(p^0 - E_{\mathbf{p}}) f(p) e^{-ip \cdot x} + \delta(p^0 + E_{\mathbf{p}}) f(p) e^{-ip \cdot x} \right) \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left(f(E_{\mathbf{p}}, \mathbf{p}) e^{-i(E_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} + f(-E_{\mathbf{p}}, \mathbf{p}) e^{-i(-E_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} \right) \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left(f(E_{\mathbf{p}}, \mathbf{p}) e^{-i(E_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} + f(-E_{\mathbf{p}}, -\mathbf{p}) e^{-i(-E_{\mathbf{p}}t - (-\mathbf{p}) \cdot \mathbf{x})} \right) \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left(f(E_{\mathbf{p}}, \mathbf{p}) e^{-i(E_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} + f(-E_{\mathbf{p}}, -\mathbf{p}) e^{i(E_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} \right) \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(a_{\mathbf{p}} e^{-ip \cdot x} + b_{\mathbf{p}}^{\dagger} e^{ip \cdot x} \right), \end{aligned}$$

where

$$a_{\mathbf{p}} = \frac{1}{\sqrt{2E_{\mathbf{p}}}} f(E_{\mathbf{p}}, \mathbf{p})$$

and

$$b_{\mathbf{p}}^{\dagger} = \frac{1}{\sqrt{2E_{\mathbf{p}}}} f(-E_{\mathbf{p}}, -\mathbf{p}).$$

In the classical theory, $a_{\mathbf{p}}$ and $b_{\mathbf{p}}$ are functions that determine the amplitude of positive-frequency and negative-frequency modes, respectively, where $\dagger$ is just the complex conjugate. In the quantum theory, $a_{\mathbf{p}}$ and $b_{\mathbf{p}}$ are promoted to operators that correspond to particles and antiparticles, where $\dagger$ is the Hermitian adjoint. If $\phi(x)$ is real-valued (or Hermitian), then $\phi^*(x) = \phi(x)$, which implies $b_{\mathbf{p}} = a_{\mathbf{p}}$ and $b_{\mathbf{p}}^{\dagger} = a_{\mathbf{p}}^{\dagger}$. Then, the above expression simplifies to the mode expansion for the real Klein-Gordon field.

(a) The conjugate momentum to $\phi(x)$, denoted $\pi(x)$, is given by the following:

$$\begin{aligned}
\pi(x) &= \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \\
&= \frac{\partial}{\partial \dot{\phi}} \left(\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi \right) \rightarrow \text{separate components and lower indices} \\
&= \frac{\partial}{\partial (\partial_0 \phi)} \left(\partial_0 \phi^* \partial_0 \phi - \partial_i \phi^* \partial_i \phi - m^2 \phi^* \phi \right) \rightarrow \text{2nd and 3rd terms give 0} \\
&= \frac{\partial (\partial_0 \phi^*)}{\partial (\partial_0 \phi)} \partial_0 \phi + \partial_0 \phi^* \frac{\partial (\partial_0 \phi)}{\partial (\partial_0 \phi)} \rightarrow \text{1st term is zero} \\
&= \partial_0 \phi^*.
\end{aligned}$$

The conjugate momentum to $\phi^*(x)$, denoted $\pi^*(x)$, is determined in a similar manner:

$$\begin{aligned}
\pi^*(x) &= \frac{\partial \mathcal{L}}{\partial \dot{\phi}^*} \\
&= \frac{\partial}{\partial \dot{\phi}^*} \left(\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi \right) \rightarrow \text{separate components and lower indices} \\
&= \frac{\partial}{\partial (\partial_0 \phi^*)} \left(\partial_0 \phi^* \partial_0 \phi - \partial_i \phi^* \partial_i \phi - m^2 \phi^* \phi \right) \rightarrow \text{2nd and 3rd terms give 0} \\
&= \frac{\partial (\partial_0 \phi^*)}{\partial (\partial_0 \phi^*)} \partial_0 \phi + \partial_0 \phi^* \frac{\partial (\partial_0 \phi)}{\partial (\partial_0 \phi^*)} \rightarrow \text{2nd term is zero} \\
&= \partial_0 \phi.
\end{aligned}$$

In order to determine the canonical commutation relations, we make use of the following commutation relations for the creation and annihilation operators:

$$\begin{aligned}
[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] &= (2\pi)^3 \delta(\mathbf{p} - \mathbf{q}), \quad [b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta(\mathbf{p} - \mathbf{q}), \\
[a_{\mathbf{p}}, a_{\mathbf{q}}] &= 0, \quad [b_{\mathbf{p}}, b_{\mathbf{q}}] = 0, \quad [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger] = 0, \quad [b_{\mathbf{p}}^\dagger, b_{\mathbf{q}}^\dagger] = 0, \\
[a_{\mathbf{p}}, b_{\mathbf{q}}] &= 0, \quad [a_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = 0, \quad [a_{\mathbf{p}}^\dagger, b_{\mathbf{q}}] = 0, \quad [a_{\mathbf{p}}^\dagger, b_{\mathbf{q}}^\dagger] = 0.
\end{aligned}$$

The canonical commutation relations at $t = 0$ are then:

$$\begin{aligned}
[\phi(\mathbf{x}), \phi(\mathbf{y})] &= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{x}} + b_{\mathbf{p}}^\dagger e^{-i\mathbf{p} \cdot \mathbf{x}}, a_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{y}} + b_{\mathbf{k}}^\dagger e^{-i\mathbf{k} \cdot \mathbf{y}} \right] \\
&= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}, a_{\mathbf{k}}] e^{i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} + [a_{\mathbf{p}}, b_{\mathbf{k}}^\dagger] e^{i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} \right. \\
&\quad \left. + [b_{\mathbf{p}}^\dagger, a_{\mathbf{k}}] e^{-i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} + [b_{\mathbf{p}}^\dagger, b_{\mathbf{k}}^\dagger] e^{-i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} \right)
\end{aligned}$$

$$= 0.$$

$$\begin{aligned}
[\pi(\mathbf{x}), \pi(\mathbf{y})] &= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} - b_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}}, a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{y}} - b_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{y}} \right] \\
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}^\dagger, a_{\mathbf{k}}^\dagger] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - [a_{\mathbf{p}}^\dagger, b_{\mathbf{k}}] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right. \\
&\quad \left. - [b_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} + [b_{\mathbf{p}}, b_{\mathbf{k}}] e^{i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
[\phi(\mathbf{x}), \pi(\mathbf{y})] &= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} + b_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}}, a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{y}} - b_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{y}} \right] \\
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - [a_{\mathbf{p}}, b_{\mathbf{k}}] e^{i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right. \\
&\quad \left. + [b_{\mathbf{p}}^\dagger, a_{\mathbf{k}}^\dagger] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - [b_{\mathbf{p}}^\dagger, b_{\mathbf{k}}] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right) \\
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left((2\pi)^3 \delta(\mathbf{p} - \mathbf{k}) e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - 0 \right. \\
&\quad \left. + 0 + (2\pi)^3 \delta(\mathbf{p} - \mathbf{k}) e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \left(e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{p}\cdot\mathbf{y}} + e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{p}\cdot\mathbf{y}} \right) \\
&= \frac{i}{2} \int \frac{d^3p}{(2\pi)^3} \left(e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right) \rightarrow \mathbf{p} \text{ to } -\mathbf{p} \text{ in 1st term} \\
&= \frac{i}{2} \int \frac{d^3p}{(2\pi)^3} \left(e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right) \\
&= i \int \frac{d^3p}{(2\pi)^3} \left(e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right) \\
&= i\delta(\mathbf{x} - \mathbf{y}).
\end{aligned}$$

$$[\phi^*(\mathbf{x}), \phi^*(\mathbf{y})] = \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} + b_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}}, a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{y}} + b_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{y}} \right]$$

$$\begin{aligned}
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}^\dagger, a_{\mathbf{k}}^\dagger] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} + [a_{\mathbf{p}}^\dagger, b_{\mathbf{k}}] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right. \\
&\quad \left. + [b_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} + [b_{\mathbf{p}}, b_{\mathbf{k}}] e^{i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
[\pi^*(\mathbf{x}), \pi^*(\mathbf{y})] &= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \left[b_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} - a_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}}, b_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{y}} - a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{y}} \right] \\
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([b_{\mathbf{p}}^\dagger, b_{\mathbf{k}}^\dagger] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - [b_{\mathbf{p}}^\dagger, a_{\mathbf{k}}] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right. \\
&\quad \left. - [a_{\mathbf{p}}, b_{\mathbf{k}}^\dagger] e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} + [a_{\mathbf{p}}, a_{\mathbf{k}}] e^{i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
[\phi^*(\mathbf{x}), \pi^*(\mathbf{y})] &= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} + b_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}}, b_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{y}} - a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{y}} \right] \\
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}^\dagger, b_{\mathbf{k}}^\dagger] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - [a_{\mathbf{p}}^\dagger, a_{\mathbf{k}}] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right. \\
&\quad \left. + [b_{\mathbf{p}}, b_{\mathbf{k}}^\dagger] e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - [b_{\mathbf{p}}, a_{\mathbf{k}}] e^{i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right) \\
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left(0 + (2\pi)^3 \delta(\mathbf{p} - \mathbf{k}) e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right. \\
&\quad \left. + (2\pi)^3 \delta(\mathbf{p} - \mathbf{k}) e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - 0 \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \left(e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{p}\cdot\mathbf{y}} + e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{p}\cdot\mathbf{y}} \right) \\
&= \frac{i}{2} \int \frac{d^3p}{(2\pi)^3} \left(e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right) \rightarrow \mathbf{p} \text{ to } -\mathbf{p} \text{ in 2nd term} \\
&= \frac{i}{2} \int \frac{d^3p}{(2\pi)^3} \left(e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right)
\end{aligned}$$

$$\begin{aligned}
&= i \int \frac{d^3 p}{(2\pi)^3} \left(e^{-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} \right) \\
&= i \delta(\mathbf{x} - \mathbf{y}).
\end{aligned}$$

$$\begin{aligned}
[\phi(\mathbf{x}), \pi^*(\mathbf{y})] &= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{x}} + b_{\mathbf{p}}^\dagger e^{-i\mathbf{p} \cdot \mathbf{x}}, b_{\mathbf{k}}^\dagger e^{-i\mathbf{k} \cdot \mathbf{y}} - a_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{y}} \right] \\
&= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}, b_{\mathbf{k}}^\dagger] e^{i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} - [a_{\mathbf{p}}, a_{\mathbf{k}}] e^{i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} \right. \\
&\quad \left. + [b_{\mathbf{p}}^\dagger, b_{\mathbf{k}}^\dagger] e^{-i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} - [b_{\mathbf{p}}^\dagger, a_{\mathbf{k}}] e^{-i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} \right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
[\phi^*(\mathbf{x}), \pi(\mathbf{y})] &= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}}^\dagger e^{-i\mathbf{p} \cdot \mathbf{x}} + b_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{x}}, a_{\mathbf{k}}^\dagger e^{-i\mathbf{k} \cdot \mathbf{y}} - b_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{y}} \right] \\
&= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}^\dagger, a_{\mathbf{k}}^\dagger] e^{-i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} - [a_{\mathbf{p}}^\dagger, b_{\mathbf{k}}] e^{-i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} \right. \\
&\quad \left. + [b_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} - [b_{\mathbf{p}}, b_{\mathbf{k}}] e^{i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} \right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
[\phi(\mathbf{x}), \phi^*(\mathbf{y})] &= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{x}} + b_{\mathbf{p}}^\dagger e^{-i\mathbf{p} \cdot \mathbf{x}}, a_{\mathbf{k}}^\dagger e^{-i\mathbf{k} \cdot \mathbf{y}} + b_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{y}} \right] \\
&= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} + [a_{\mathbf{p}}, b_{\mathbf{k}}] e^{i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} \right. \\
&\quad \left. + [b_{\mathbf{p}}^\dagger, a_{\mathbf{k}}^\dagger] e^{-i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} + [b_{\mathbf{p}}^\dagger, b_{\mathbf{k}}] e^{-i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} \right) \\
&= \int \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left((2\pi)^3 \delta(\mathbf{p} - \mathbf{k}) e^{i\mathbf{p} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{y}} + 0 \right. \\
&\quad \left. + 0 - (2\pi)^3 \delta(\mathbf{p} - \mathbf{k}) e^{-i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{y}} \right)
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{p}\cdot\mathbf{y}} - e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{p}\cdot\mathbf{y}} \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left(e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} - e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right) \rightarrow \mathbf{p} \text{ to } -\mathbf{p} \text{ in 1st term} \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left(e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} - e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
[\pi(\mathbf{x}), \pi^*(\mathbf{y})] &= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \left[a_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} - b_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}}, b_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{y}} - a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{y}} \right] \\
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left([a_{\mathbf{p}}^\dagger, b_{\mathbf{k}}^\dagger] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} - [a_{\mathbf{p}}^\dagger, a_{\mathbf{k}}] e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right. \\
&\quad \left. - [b_{\mathbf{p}}, b_{\mathbf{k}}^\dagger] e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} + [b_{\mathbf{p}}, a_{\mathbf{k}}] e^{i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right) \\
&= \int \int \frac{d^3p}{(2\pi)^3} \frac{d^3k}{(2\pi)^3} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{k}})}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left(0 + (2\pi)^3 \delta(\mathbf{p} - \mathbf{k}) e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{k}\cdot\mathbf{y}} \right. \\
&\quad \left. - (2\pi)^3 \delta(\mathbf{p} - \mathbf{k}) e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{k}\cdot\mathbf{y}} + 0 \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \frac{(iE_{\mathbf{p}})}{\sqrt{2E_{\mathbf{p}}}} \left(e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{p}\cdot\mathbf{y}} - e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{p}\cdot\mathbf{y}} \right) \\
&= -\frac{1}{2} \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left(e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} - e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right) \rightarrow \mathbf{p} \text{ to } -\mathbf{p} \text{ in 2nd term} \\
&= -\frac{1}{2} \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left(e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} - e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right) \\
&= 0.
\end{aligned}$$

Note that it is sufficient to impose the canonical commutation relations at $t = 0$, as they are automatically satisfied at a later time, t . This can be shown by making ϕ and π time-dependent in the Heisenberg picture. The key idea here is outlined as follows:

$$[A(\mathbf{x}, t), B(\mathbf{y}, t)] = [e^{iHt} A(\mathbf{x}, 0) e^{-iHt}, e^{iHt} B(\mathbf{y}, 0) e^{-iHt}] = e^{iHt} [A(\mathbf{x}, 0), B(\mathbf{y}, 0)] e^{-iHt}.$$

The Hamiltonian is:

$$H = \int d^3x \left(\sum_i \pi_i(\mathbf{x}) \dot{\phi}_i(\mathbf{x}) - \mathcal{L} \right)$$

$$\begin{aligned}
H &= \int d^3x \left(\pi(\mathbf{x}) \dot{\phi}(\mathbf{x}) + \pi^*(\mathbf{x}) \dot{\phi}^*(\mathbf{x}) - \partial_\mu \phi^* \partial^\mu \phi + m^2 \phi^* \phi \right) \\
&= \int d^3x \left(\pi(\mathbf{x}) \pi^*(\mathbf{x}) + \pi^*(\mathbf{x}) \pi(\mathbf{x}) - \partial_\mu \phi^* \partial^\mu \phi + m^2 \phi^* \phi \right) \\
&= \int d^3x \left(\pi(\mathbf{x}) \pi^*(\mathbf{x}) + \pi^*(\mathbf{x}) \pi(\mathbf{x}) - \partial_0 \phi^* \partial^0 \phi + \partial_i \phi^* \partial^i \phi + m^2 \phi^* \phi \right) \\
&= \int d^3x \left(\pi(\mathbf{x}) \pi^*(\mathbf{x}) + \pi^*(\mathbf{x}) \pi(\mathbf{x}) - \pi(\mathbf{x}) \pi^*(\mathbf{x}) + \partial_i \phi^* \partial^i \phi + m^2 \phi^* \phi \right) \\
&= \int d^3x \left(\pi^*(\mathbf{x}) \pi(\mathbf{x}) + \nabla \phi^* \cdot \nabla \phi + m^2 \phi^* \phi \right).
\end{aligned}$$

The Heisenberg equation of motion for $\phi(x)$ is:

$$\begin{aligned}
i \frac{\partial}{\partial t} \mathcal{O} &= [\mathcal{O}, H] \\
i \frac{\partial}{\partial t} \phi(x) &= \int d^3x \left[\phi(x), \left(\pi^*(y) \pi(y) + \nabla \phi^*(y) \cdot \nabla \phi(y) + m^2 \phi^*(y) \phi(y) \right) \right] \\
&= \int d^3x \left([\phi(x), \pi^*(y) \pi(y)] + [\phi(x), \partial_i \phi^*(y) \partial^i \phi(y)] + m^2 [\phi(x), \phi^*(y) \phi(y)] \right) \\
&= \int d^3x \left([\phi(x), \pi^*(y)] \pi(y) + \pi^*(y) [\phi(x), \pi(y)] + [\phi(x), \partial_i \phi^*(y)] \partial^i \phi(y) \right. \\
&\quad \left. + \partial_i \phi^*(y) [\phi(x), \partial^i \phi(y)] + m^2 [\phi(x), \phi^*(y)] \phi(y) + m^2 \phi^*(y) [\phi(x), \phi(y)] \right) \\
&= \int d^3x \left(0 + \pi^*(y) (i \delta(\mathbf{x} - \mathbf{y})) + 0 + 0 + 0 + 0 \right) \\
&= i \pi^*(x),
\end{aligned}$$

which is just the conjugate momentum expression that was previously calculated. The Heisenberg equation of motion for $\pi^*(x)$ is:

$$\begin{aligned}
i \frac{\partial}{\partial t} \mathcal{O} &= [\mathcal{O}, H] \\
i \frac{\partial}{\partial t} \pi^*(x) &= \left[\pi^*(x), \int d^3y \left(\pi^*(y) \pi(y) + \nabla \phi^*(y) \cdot \nabla \phi(y) + m^2 \phi^*(y) \phi(y) \right) \right] \\
&= \int d^3y \left[\pi^*(x), \pi^*(y) \pi(y) \right] \\
&\quad + \left[\pi^*(x), \int d^3y \nabla \phi^*(y) \cdot \nabla \phi(y) \right] \\
&\quad + m^2 \int d^3y \left[\pi^*(x), \phi^*(y) \phi(y) \right] \\
&= \int d^3y \left([\pi^*(x), \pi^*(y)] \pi(y) + \pi^*(y) [\pi^*(x), \pi(y)] \right)
\end{aligned}$$

$$\begin{aligned}
& + \left[\pi^*(x), \int d^3y \left(\nabla \cdot (\phi^*(y) \nabla \phi(y)) - \phi^*(y) \nabla^2 \phi(y) \right) \right] \\
& + m^2 \int d^3y \left([\pi^*(x), \phi^*(y)] \phi(y) + \phi^*(y) [\pi^*(x), \phi(y)] \right) \\
& = 0 + 0 \\
& + \left[\pi^*(x), \int dS \left((\phi^*(y) \nabla \phi(y)) \cdot \hat{\mathbf{n}} \right) - \int d^3y \phi^*(y) \nabla^2 \phi(y) \right] \\
& + m^2 \int d^3y \left(-i\delta(\mathbf{x} - \mathbf{y}) \phi(y) \right) + 0 \\
& = 0 - \int d^3y \left[\pi^*(x), \phi^*(y) \nabla^2 \phi(y) \right] - im^2 \phi(x) \\
& = - \int d^3y \left(\phi^*(y) [\pi^*(x), \nabla^2 \phi(y)] + [\pi^*(x), \phi^*(y)] \nabla^2 \phi(y) \right) - im^2 \phi(x) \\
& = - \int d^3y \left(\phi^*(y) (\nabla^2 [\phi^*(x), \phi(y)]) - i\delta(\mathbf{x} - \mathbf{y}) \nabla^2 \phi(y) \right) - im^2 \phi(x) \\
& = \int d^3y \left(i\delta(\mathbf{x} - \mathbf{y}) \nabla^2 \phi(y) \right) - im^2 \phi(x) \\
& = i(\nabla^2 \phi(x) - m^2 \phi(x)) \\
& = i(\nabla^2 - m^2) \phi(x),
\end{aligned}$$

where the vector identity $\nabla \cdot (A \nabla B) = A \nabla^2 B + \nabla A \cdot \nabla B$ and Gauss' theorem were used to handle the gradient terms (the surface term is zero). We can show these equations, when combined, are indeed the Klein-Gordon equation as follows:

$$\begin{aligned}
i \frac{\partial}{\partial t} \pi^*(x) &= i(\nabla^2 - m^2) \phi(x) \\
\frac{\partial}{\partial t} \left(\frac{\partial}{\partial t} \phi(x) \right) &= (\nabla^2 - m^2) \phi(x) \\
\frac{\partial^2}{\partial t^2} \phi(x) &= (\nabla^2 - m^2) \phi(x) \\
\partial_0 \partial^0 \phi(x) &= \partial_i \partial^i \phi(x) - m^2 \phi(x) \\
\partial_0 \partial^0 \phi(x) - \partial_i \partial^i \phi(x) + m^2 \phi(x) &= 0 \\
\partial_\mu \partial^\mu \phi(x) + m^2 \phi(x) &= 0 \\
\left(\partial_\mu \partial^\mu + m^2 \right) \phi(x) &= 0 \rightarrow \text{Klein-Gordon equation.}
\end{aligned}$$

The Klein-Gordon equation for $\phi^*(x)$ and $\pi(x)$ can be found in the same manner.

(b) The Hamiltonian, H , can be rewritten in terms of creation and annihilation operators as follows:

$$\begin{aligned}
H &= \int d^3x \left(\pi^*(\mathbf{x})\pi(\mathbf{x}) + \nabla\phi^*(\mathbf{x}) \cdot \nabla\phi(\mathbf{x}) + m^2\phi^*(\mathbf{x})\phi(\mathbf{x}) \right) \\
&= \int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \\
&\quad \times \left(\frac{iE_{\mathbf{p}}}{\sqrt{2E_{\mathbf{p}}}} \frac{iE_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}} (b_{\mathbf{p}}^\dagger e^{ip \cdot x} - a_{\mathbf{p}} e^{-ip \cdot x}) (a_{\mathbf{k}}^\dagger e^{ik \cdot x} - b_{\mathbf{k}} e^{-ik \cdot x}) \right. \\
&\quad + \frac{(-i\mathbf{p})}{\sqrt{2E_{\mathbf{p}}}} \cdot \frac{(i\mathbf{k})}{\sqrt{2E_{\mathbf{k}}}} (a_{\mathbf{p}}^\dagger e^{ip \cdot x} - b_{\mathbf{p}} e^{-ip \cdot x}) (a_{\mathbf{k}} e^{-ik \cdot x} - b_{\mathbf{k}}^\dagger e^{ik \cdot x}) \\
&\quad \left. + m^2 \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} (a_{\mathbf{p}}^\dagger e^{ip \cdot x} + b_{\mathbf{p}} e^{-ip \cdot x}) (a_{\mathbf{k}} e^{-ik \cdot x} + b_{\mathbf{k}}^\dagger e^{ik \cdot x}) \right) \\
&= \int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left(-E_{\mathbf{p}} E_{\mathbf{k}} (b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger e^{i(p+k) \cdot x} - b_{\mathbf{p}}^\dagger b_{\mathbf{k}} e^{i(p-k) \cdot x} - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger e^{-i(p-k) \cdot x} + a_{\mathbf{p}} b_{\mathbf{k}} e^{-i(p+k) \cdot x}) \right. \\
&\quad + \mathbf{p} \cdot \mathbf{k} (a_{\mathbf{p}}^\dagger a_{\mathbf{k}} e^{i(p-k) \cdot x} - a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger e^{i(p+k) \cdot x} - b_{\mathbf{p}} a_{\mathbf{k}} e^{-i(p+k) \cdot x} + b_{\mathbf{p}} b_{\mathbf{k}}^\dagger e^{-i(p-k) \cdot x}) \\
&\quad \left. + m^2 (a_{\mathbf{p}}^\dagger a_{\mathbf{k}} e^{i(p-k) \cdot x} + a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger e^{i(p+k) \cdot x} + b_{\mathbf{p}} a_{\mathbf{k}} e^{-i(p+k) \cdot x} + b_{\mathbf{p}} b_{\mathbf{k}}^\dagger e^{-i(p-k) \cdot x}) \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} (2\pi)^3 \\
&\quad \times \left(-E_{\mathbf{p}} E_{\mathbf{k}} (b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger \delta(\mathbf{p} + \mathbf{k}) - b_{\mathbf{p}}^\dagger b_{\mathbf{k}} \delta(\mathbf{p} - \mathbf{k}) - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger \delta(\mathbf{p} - \mathbf{k}) + a_{\mathbf{p}} b_{\mathbf{k}} \delta(\mathbf{p} + \mathbf{k})) \right. \\
&\quad + \mathbf{p} \cdot \mathbf{k} (a_{\mathbf{p}}^\dagger a_{\mathbf{k}} \delta(\mathbf{p} - \mathbf{k}) - a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger \delta(\mathbf{p} + \mathbf{k}) - b_{\mathbf{p}} a_{\mathbf{k}} \delta(\mathbf{p} + \mathbf{k}) + b_{\mathbf{p}} b_{\mathbf{k}}^\dagger \delta(\mathbf{p} - \mathbf{k})) \\
&\quad \left. + m^2 (a_{\mathbf{p}}^\dagger a_{\mathbf{k}} \delta(\mathbf{p} - \mathbf{k}) + a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger \delta(\mathbf{p} + \mathbf{k}) + b_{\mathbf{p}} a_{\mathbf{k}} \delta(\mathbf{p} + \mathbf{k}) + b_{\mathbf{p}} b_{\mathbf{k}}^\dagger \delta(\mathbf{p} - \mathbf{k})) \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \\
&\quad \times \left(-E_{\mathbf{p}}^2 b_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger + E_{\mathbf{p}}^2 b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + E_{\mathbf{p}}^2 a_{\mathbf{p}} a_{\mathbf{p}}^\dagger - E_{\mathbf{p}}^2 a_{\mathbf{p}} b_{-\mathbf{p}} \right. \\
&\quad + \mathbf{p}^2 a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + \mathbf{p}^2 a_{\mathbf{p}}^\dagger b_{-\mathbf{p}}^\dagger + \mathbf{p}^2 b_{\mathbf{p}} a_{-\mathbf{p}} + \mathbf{p}^2 b_{\mathbf{p}} b_{\mathbf{p}}^\dagger \\
&\quad \left. + m^2 a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + m^2 a_{\mathbf{p}}^\dagger b_{-\mathbf{p}}^\dagger + m^2 b_{\mathbf{p}} a_{-\mathbf{p}} + m^2 b_{\mathbf{p}} b_{\mathbf{p}}^\dagger \right)
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \\
&\quad \times \left((-E_{\mathbf{p}}^2 + \mathbf{p}^2 + m^2) b_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger + (-E_{\mathbf{p}}^2 + \mathbf{p}^2 + m^2) a_{\mathbf{p}} b_{-\mathbf{p}} \right. \\
&\quad \left. + E_{\mathbf{p}}^2 a_{\mathbf{p}} a_{\mathbf{p}}^\dagger + \mathbf{p}^2 a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + m^2 a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + E_{\mathbf{p}}^2 b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + \mathbf{p}^2 b_{\mathbf{p}} b_{\mathbf{p}}^\dagger + m^2 b_{\mathbf{p}} b_{\mathbf{p}}^\dagger \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left(E_{\mathbf{p}}^2 (a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + (2\pi)^3 \delta(0)) + \mathbf{p}^2 a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + m^2 a_{\mathbf{p}}^\dagger a_{\mathbf{p}} \right. \\
&\quad \left. + E_{\mathbf{p}}^2 b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + \mathbf{p}^2 (b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + (2\pi)^3 \delta(0)) + m^2 (b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + (2\pi)^3 \delta(0)) \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left((E_{\mathbf{p}}^2 + \mathbf{p}^2 + m^2) a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + E_{\mathbf{p}}^2 (2\pi)^3 \delta(0) \right. \\
&\quad \left. + (E_{\mathbf{p}}^2 + \mathbf{p}^2 + m^2) b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + \mathbf{p}^2 (2\pi)^3 \delta(0) + m^2 (2\pi)^3 \delta(0) \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left(2E_{\mathbf{p}}^2 a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + 2E_{\mathbf{p}}^2 b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + 2E_{\mathbf{p}}^2 (2\pi)^3 \delta(0) \right) \\
&= \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + (2\pi)^3 \delta(0) \right) \\
&= \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + b_{\mathbf{p}}^\dagger b_{\mathbf{p}} \right),
\end{aligned}$$

where the $\delta(0)$ term was dropped in the last step. To show that this theory contains two sets of particles with mass m , we can interpret the Hamiltonian in terms of its action on the one-particle states $|\mathbf{p}\rangle_a = a_{\mathbf{p}}^\dagger |0\rangle$ and $|\mathbf{p}\rangle_b = b_{\mathbf{p}}^\dagger |0\rangle$, where the state $|0\rangle$ is the ground state, as follows:

$$\begin{aligned}
H |\mathbf{p}\rangle_a &= \int \frac{d^3q}{(2\pi)^3} E_{\mathbf{q}} \left(a_{\mathbf{q}}^\dagger a_{\mathbf{q}} + b_{\mathbf{q}}^\dagger b_{\mathbf{q}} \right) a_{\mathbf{p}}^\dagger |0\rangle \\
&= \int \frac{d^3q}{(2\pi)^3} E_{\mathbf{q}} \left(a_{\mathbf{q}}^\dagger a_{\mathbf{q}} a_{\mathbf{p}}^\dagger |0\rangle + b_{\mathbf{q}}^\dagger b_{\mathbf{q}} a_{\mathbf{p}}^\dagger |0\rangle \right) \\
&= \int \frac{d^3q}{(2\pi)^3} E_{\mathbf{q}} \left((2\pi)^3 \delta(\mathbf{q} - \mathbf{p}) a_{\mathbf{p}}^\dagger |0\rangle + b_{\mathbf{q}}^\dagger a_{\mathbf{p}}^\dagger b_{\mathbf{q}} |0\rangle \right) \\
&= E_{\mathbf{p}} a_{\mathbf{p}}^\dagger |0\rangle \\
&= E_{\mathbf{p}} |\mathbf{p}\rangle_a.
\end{aligned}$$

Similarly,

$$H |\mathbf{p}\rangle_b = E_{\mathbf{p}} |\mathbf{p}\rangle_b.$$

Therefore, the Hamiltonian describes two sets of particles: one created by $a_{\mathbf{p}}^\dagger$ and the other by $b_{\mathbf{p}}^\dagger$, each with momentum $\mathbf{p}$ and energy $E_{\mathbf{p}}$. Since $E_{\mathbf{p}}^2 = p^2 + m^2$, both particles have the same mass m .

(c) The conserved charge, Q , can be rewritten in terms of creation and annihilation operators as follows:

$$\begin{aligned}
Q &= \int d^3x \frac{i}{2} \left(\phi^*(\mathbf{x}) \pi^*(\mathbf{x}) - \pi(\mathbf{x}) \phi(\mathbf{x}) \right) \\
&= \int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{i}{2} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left((iE_{\mathbf{k}}) (a_{\mathbf{p}}^\dagger e^{ip \cdot x} + b_{\mathbf{p}} e^{-ip \cdot x}) (b_{\mathbf{k}}^\dagger e^{ik \cdot x} - a_{\mathbf{k}} e^{-ik \cdot x}) \right. \\
&\quad \left. - (iE_{\mathbf{p}}) (a_{\mathbf{p}}^\dagger e^{ip \cdot x} - b_{\mathbf{p}} e^{-ip \cdot x}) (a_{\mathbf{k}} e^{-ik \cdot x} + b_{\mathbf{k}}^\dagger e^{ik \cdot x}) \right) \\
&= \int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{i}{2} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left((iE_{\mathbf{k}}) (a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger e^{ip \cdot x} e^{ik \cdot x} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}} e^{ip \cdot x} e^{-ik \cdot x} + b_{\mathbf{p}} b_{\mathbf{k}}^\dagger e^{-ip \cdot x} e^{ik \cdot x} - b_{\mathbf{p}} a_{\mathbf{k}} e^{-ip \cdot x} e^{-ik \cdot x}) \right. \\
&\quad \left. - (iE_{\mathbf{p}}) (a_{\mathbf{p}}^\dagger a_{\mathbf{k}} e^{ip \cdot x} e^{-ik \cdot x} + a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger e^{ip \cdot x} e^{ik \cdot x} - b_{\mathbf{p}} a_{\mathbf{k}} e^{-ip \cdot x} e^{-ik \cdot x} - b_{\mathbf{p}} b_{\mathbf{k}}^\dagger e^{-ip \cdot x} e^{ik \cdot x}) \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{i}{2} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \\
&\quad \times \left((iE_{\mathbf{k}}) (a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger \delta(\mathbf{p} + \mathbf{k}) - a_{\mathbf{p}}^\dagger a_{\mathbf{k}} \delta(\mathbf{p} - \mathbf{k}) + b_{\mathbf{p}} b_{\mathbf{k}}^\dagger \delta(\mathbf{p} - \mathbf{k}) - b_{\mathbf{p}} a_{\mathbf{k}} \delta(\mathbf{p} + \mathbf{k})) \right. \\
&\quad \left. - (iE_{\mathbf{p}}) (a_{\mathbf{p}}^\dagger a_{\mathbf{k}} \delta(\mathbf{p} - \mathbf{k}) + a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger \delta(\mathbf{p} + \mathbf{k}) - b_{\mathbf{p}} a_{\mathbf{k}} \delta(\mathbf{p} + \mathbf{k}) - b_{\mathbf{p}} b_{\mathbf{k}}^\dagger \delta(\mathbf{p} - \mathbf{k})) \right) \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{i}{2} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \\
&\quad \times \left((iE_{\mathbf{p}}) (a_{\mathbf{p}}^\dagger b_{-\mathbf{p}}^\dagger - a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + b_{\mathbf{p}} b_{\mathbf{p}}^\dagger - b_{\mathbf{p}} a_{-\mathbf{p}}) \right. \\
&\quad \left. - (iE_{\mathbf{p}}) (a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + a_{\mathbf{p}}^\dagger b_{-\mathbf{p}}^\dagger - b_{\mathbf{p}} a_{-\mathbf{p}} - b_{\mathbf{p}} b_{\mathbf{p}}^\dagger) \right) \\
&= \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} (a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - b_{\mathbf{p}} b_{\mathbf{p}}^\dagger) \\
&= \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} (a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - b_{\mathbf{p}}^\dagger b_{\mathbf{p}} - (2\pi)^3 \delta(0))
\end{aligned}$$

$$= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} (a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - b_{\mathbf{p}}^\dagger b_{\mathbf{p}}),$$

where the $\delta(0)$ term was dropped in the last step. To evaluate the charge carried by each particle type, we can interpret the operator Q in terms of its action on the one-particle states $|\mathbf{p}\rangle_a = a_{\mathbf{p}}^\dagger |0\rangle$ and $|\mathbf{p}\rangle_b = b_{\mathbf{p}}^\dagger |0\rangle$, where the state $|0\rangle$ is the ground state, as follows:

$$\begin{aligned} Q |\mathbf{p}\rangle_a &= \frac{1}{2} \int \frac{d^3 q}{(2\pi)^3} (a_{\mathbf{q}}^\dagger a_{\mathbf{q}} - b_{\mathbf{q}}^\dagger b_{\mathbf{q}}) a_{\mathbf{p}}^\dagger |0\rangle \\ &= \frac{1}{2} \int \frac{d^3 q}{(2\pi)^3} (a_{\mathbf{q}}^\dagger a_{\mathbf{q}} a_{\mathbf{p}}^\dagger |0\rangle - b_{\mathbf{q}}^\dagger b_{\mathbf{q}} a_{\mathbf{p}}^\dagger |0\rangle) \\ &= \frac{1}{2} \int \frac{d^3 q}{(2\pi)^3} ((2\pi)^3 \delta(\mathbf{q} - \mathbf{p}) a_{\mathbf{q}}^\dagger |0\rangle - b_{\mathbf{q}}^\dagger a_{\mathbf{p}}^\dagger b_{\mathbf{q}} |0\rangle) \\ &= \frac{1}{2} a_{\mathbf{p}}^\dagger |0\rangle \\ &= \frac{1}{2} |\mathbf{p}\rangle_a. \end{aligned}$$

Similarly,

$$Q |\mathbf{p}\rangle_b = -\frac{1}{2} |\mathbf{p}\rangle_b.$$

Therefore, the particles created by $a_{\mathbf{p}}^\dagger$ carry a charge of $+1/2$, while the particles created by $b_{\mathbf{p}}^\dagger$ carry a charge of $-1/2$.

(d) For the case of two complex Klein-Gordon fields with the same mass, labeled as $\phi_a(x)$, where $a = 1, 2$, the action can be written as:

$$S = \int d^4 x (\partial_\mu \phi_a^* \partial^\mu \phi_a - m^2 \phi_a^* \phi_a).$$

To identify the conserved charges, we first need to determine the continuous transformations that leave the Lagrangian invariant. These transformations are of the form $\phi_a \rightarrow U_{ab} \phi_b$, where U_{ab} are components of a linear operator. Since we have two fields, transformations satisfying these conditions are unitary transformations described by the unitary group $U(2)$, which is the set of all 2×2 unitary matrices. This set forms a non-Abelian group with the group operation being matrix multiplication. It is important to note that U_{ab} are the elements of these matrices. To analyze the symmetry structure of this theory, we make use of the generators associated with the $U(1)$ and $SU(2)$ subgroups of $U(2)$. The $U(1)$ subgroup has one generator, denoted T^0 , given by the identity matrix multiplied by i , i.e., $T^0 = i\mathbb{I}$. The Lagrangian is invariant under the transformation:

$$\phi_a \rightarrow (e^{-\alpha T^0})_{ab} \phi_b,$$

where α is an infinitesimal parameter. Furthermore, this transformation gives the following infinitesimal change:

$$(e^{-\alpha T^0})_{ab}\phi_b = (\mathbb{I} - i\alpha\mathbb{I})_{ab}\phi_b + \mathcal{O}(\alpha^2) \approx \phi_a - i\alpha\phi_a.$$

For U(1), we have the following Noether current:

$$\begin{aligned} j^\mu(x) &= \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\Delta\phi_a + \Delta\phi_a^*\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a^*)} \\ &= (\partial_\mu\phi_a^*)(-i\alpha\phi_a) + (i\alpha\phi_a^*)(\partial_\mu\phi_a) \\ &= i(\phi_a^*(\partial_\mu\phi_a) - (\partial_\mu\phi_a^*)\phi_a), \end{aligned}$$

where we have arbitrarily chosen α . Accordingly, the single charge from this symmetry, which is the generalization of part (c), is given by:

$$\begin{aligned} Q &= \int j^0 d^3x \\ &= i \int d^3x (\phi_a^* \dot{\phi}_a - \dot{\phi}_a^* \phi_a) \rightarrow \text{use conjugate momenta notation} \\ &= i \int d^3x (\phi_a^* \pi_a^* - \pi_a \phi_a). \end{aligned}$$

The U(2) subgroup has three generators, which are the Pauli matrices, σ^k , multiplied by $i/2$, i.e., $T^k = \frac{i}{2}\sigma^k$ for $k = 1, 2, 3$. The Lagrangian is invariant under the transformation:

$$\phi_a \rightarrow (e^{-\beta^k T^k})_{ab}\phi_b,$$

where β^k is an infinitesimal parameter. Similarly, this transformation gives the following infinitesimal change:

$$(e^{-\beta^k T^k})_{ab}\phi_b = (\mathbb{I} - \frac{i}{2}\beta^k\sigma^k)_{ab}\phi_b + \mathcal{O}((\beta^k)^2) \approx \phi_a - \frac{i}{2}(\beta^k\sigma^k)_{ab}\phi_b.$$

For SU(2), we have the following Noether currents:

$$\begin{aligned} j^{\mu,k}(x) &= \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\Delta^k\phi_a + \Delta^k\phi_a^*\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a^*)} \\ &= \left(\partial_\mu\phi_a^*\right)\left(-\frac{i}{2}(\beta^k\sigma^k)_{ab}\phi_b\right) + \left(\frac{i}{2}(\beta^k\sigma^k)^*_{ab}\phi_b^*\right)\left(\partial_\mu\phi_a\right) \\ &= \frac{i}{2}\left((\sigma^k)^*_{ab}\phi_b^*(\partial_\mu\phi_a) - (\partial_\mu\phi_a^*)(\sigma^k)_{ab}\phi_b\right) \rightarrow (\sigma^k)^*_{ab} = (\sigma^k)_{ba} \text{ and relabel indices} \\ &= \frac{i}{2}\left((\sigma^k)_{ab}\phi_a^*(\partial_\mu\phi_b) - (\partial_\mu\phi_a^*)(\sigma^k)_{ab}\phi_b\right) \\ &= \frac{i}{2}\left(\phi_a^*(\sigma^k)_{ab}(\partial_\mu\phi_b) - (\partial_\mu\phi_a^*)(\sigma^k)_{ab}\phi_b\right) \end{aligned}$$

where we have arbitrarily chosen β^k . Accordingly, the three charges from this symmetry are given by:

$$\begin{aligned}
Q^k &= \int j^{0,k} d^3x \\
&= \frac{i}{2} \int d^3x \left(\phi_a^*(\sigma^k)_{ab} \dot{\phi}_b - \dot{\phi}_a^*(\sigma^k)_{ab} \phi_b \right) \rightarrow \text{use conjugate momenta notation} \\
&= \frac{i}{2} \int d^3x \left(\phi_a^*(\sigma^k)_{ab} \pi_b^* - \pi_a(\sigma^k)_{ab} \phi_b \right).
\end{aligned}$$

These charges having the commutation relations of angular momentum is shown as follows:

$$\begin{aligned}
[Q^i, Q^j] &= -\frac{1}{4} \int d^3x \int d^3y \left[\left(\phi_a^*(\sigma^i)_{ab} \pi_b^* - \pi_a(\sigma^i)_{ab} \phi_b \right), \left(\phi_c^*(\sigma^j)_{cd} \pi_d^* - \pi_c(\sigma^j)_{cd} \phi_d \right) \right] \\
&= -\frac{1}{4} \int d^3x \int d^3y \\
&\quad \times \left(\left[\phi_a^*(\sigma^i)_{ab} \pi_b^*, \phi_c^*(\sigma^j)_{cd} \pi_d^* \right] - \left[\phi_a^*(\sigma^i)_{ab} \pi_b^*, \pi_c(\sigma^j)_{cd} \phi_d \right] \right. \\
&\quad \left. - \left[\pi_a(\sigma^i)_{ab} \phi_b, \phi_c^*(\sigma^j)_{cd} \pi_d^* \right] + \left[\pi_a(\sigma^i)_{ab} \phi_b, \pi_c(\sigma^j)_{cd} \phi_d \right] \right) \\
&= -\frac{1}{4} \int d^3x \int d^3y (\sigma^i)_{ab} (\sigma^j)_{cd} \\
&\quad \times \left(\left[\phi_a^* \pi_b^*, \phi_c^* \pi_d^* \right] - \left[\phi_a^* \pi_b^*, \pi_c \phi_d \right] - \left[\pi_a \phi_b, \phi_c^* \pi_d^* \right] + \left[\pi_a \phi_b, \pi_c \phi_d \right] \right) \\
&= -\frac{1}{4} \int d^3x \int d^3y (\sigma^i)_{ab} (\sigma^j)_{cd} \\
&\quad \times \left(\phi_a^* \left[\pi_b^*, \phi_c^* \right] \pi_d^* + \left[\phi_a^*, \phi_c^* \right] \pi_b^* \pi_d^* + \left[\phi_c^*, \phi_a^* \right] \pi_b^* \pi_d^* + \phi_c^* \left[\phi_a^*, \pi_d^* \right] \pi_b^* \right. \\
&\quad - \phi_a^* \left[\pi_b^*, \pi_c \right] \phi_d - \left[\phi_a^*, \pi_c \right] \pi_b^* \phi_d - \pi_c \phi_a^* \left[\pi_b^*, \phi_d \right] - \pi_c \left[\phi_a^*, \phi_d \right] \pi_b^* \\
&\quad - \pi_a \left[\phi_b, \phi_c^* \right] \pi_d^* - \left[\pi_a, \phi_c^* \right] \phi_b \pi_d^* - \phi_c^* \pi_a \left[\phi_b, \pi_d^* \right] - \phi_c^* \left[\pi_a, \pi_d^* \right] \phi_b \\
&\quad \left. + \pi_a \left[\phi_b, \pi_c \right] \phi_d + \left[\pi_a, \pi_c \right] \phi_b \phi_d + \pi_c \pi_a \left[\phi_b, \phi_d \right] + \pi_c \left[\pi_a, \phi_d \right] \phi_b \right) \\
&= -\frac{1}{4} \int d^3x \int d^3y (\sigma^i)_{ab} (\sigma^j)_{cd} \\
&\quad \times \left(\phi_a^* \left(-i \delta^{bc} \delta^{(3)}(\mathbf{x} - \mathbf{y}) \right) \pi_d^* + 0 + 0 + \phi_c^* \left(i \delta^{ad} \delta^{(3)}(\mathbf{x} - \mathbf{y}) \right) \pi_b^* \right. \\
&\quad - 0 - 0 - 0 - 0 \\
&\quad \left. - 0 - 0 - 0 - 0 \right)
\end{aligned}$$

$$\begin{aligned}
& + \pi_a (i\delta^{bc}\delta^{(3)}(\mathbf{x} - \mathbf{y}))\phi_d + 0 + 0 + \pi_c (-i\delta^{ad}\delta^{(3)}(\mathbf{x} - \mathbf{y}))\phi_b \Big) \\
& = -\frac{1}{4} \int d^3x (\sigma^i)_{ab}(\sigma^j)_{cd} \left(-i\delta^{bc}\phi_a^*\pi_d^* + i\delta^{ad}\phi_c^*\pi_b^* + i\delta^{bc}\pi_a\phi_d - i\delta^{ad}\pi_c\phi_b \right) \\
& = \frac{i}{4} \int d^3x \left(\phi_a^*(\sigma^i\sigma^j)_{ad}\pi_d^* - \phi_c^*(\sigma^j\sigma^i)_{cb}\pi_b^* - \pi_a(\sigma^i\sigma^j)_{ad}\phi_d + \pi_c(\sigma^j\sigma^i)_{cd}\phi_b \right) \\
& = \frac{i}{4} \int d^3x \left(\phi_a^*(\sigma^i\sigma^j)_{ab}\pi_b^* - \phi_a^*(\sigma^j\sigma^i)_{ab}\pi_b^* - \pi_a(\sigma^i\sigma^j)_{ab}\phi_b + \pi_a(\sigma^j\sigma^i)_{ab}\phi_b \right) \\
& = \frac{i}{4} \int d^3x \left(\phi_a^* \left((\sigma^i\sigma^j)_{ab} - (\sigma^j\sigma^i)_{ab} \right) \pi_b^* - \pi_a \left((\sigma^i\sigma^j)_{ab} - (\sigma^j\sigma^i)_{ab} \right) \phi_b \right) \\
& = \frac{i}{4} \int d^3x \left(\phi_a^* [\sigma^i, \sigma^j]_{ab} \pi_b^* - \pi_a [\sigma^i, \sigma^j]_{ab} \phi_b \right) \\
& = \frac{i}{4} \int d^3x \left(\phi_a^* \left(2i\epsilon^{ijk}(\sigma^k)_{ab} \right) \pi_b^* - \pi_a \left(2i\epsilon^{ijk}(\sigma^k)_{ab} \right) \phi_b \right) \\
& = i\epsilon^{ijk} \int d^3x \frac{i}{2} \left(\phi_a^*(\sigma^k)_{ab}\pi_b^* - \pi_a(\sigma^k)_{ab}\phi_b \right) \\
& = i\epsilon^{ijk} Q^k.
\end{aligned}$$

These results can be naturally generalized to the case of n identical complex scalar fields. In this context, the relevant symmetry group becomes the unitary group $U(n)$. This group has n^2 generators: one corresponding to $U(1)$ and $n^2 - 1$ corresponding to $SU(n)$. The associated Noether currents, $j^{\mu,k}$, and conserved charges, Q^k , have the same form as above. Additionally, the Pauli matrices are replaced by the generators T^k of $SU(n)$, and the Levi-Civita symbol is replaced by the structure constants f^{ijk} , which define the commutation relations between the generators. To address the footnote, you can treat the n complex scalar fields by their real and imaginary components as $2n$ real fields. Then, instead of transformations described by the unitary group $U(n)$, we consider transformations described by the special orthogonal group $SO(N)$, which has $N(N-1)/2$ generators, or $n(2n-1)$ generators with $N = 2n$.

2.3 Spacelike propagation amplitude

Evaluate the function

$$\langle 0 | \phi(x)\phi(y) | 0 \rangle = D(x-y) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} e^{-ip \cdot (x-y)},$$

for $(x-y)$ spacelike so that $(x-y)^2 = -r^2$, explicitly in terms of Bessel functions. For the case where $(x-y)$ is spacelike, we have $x^0 - y^0 = 0$ and $\mathbf{x} - \mathbf{y} = \mathbf{r}$. Using this, the propagation amplitude becomes:

$$D(x-y) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} e^{-ip \cdot (x-y)} \rightarrow \text{apply spacelike condition}$$

$$\begin{aligned}
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} e^{i\mathbf{p}\cdot\mathbf{r}} \rightarrow \text{switch to spherical coordinates and rewrite } E_{\mathbf{p}} \\
&= \frac{1}{(2\pi)^3} \int_0^\infty p^2 dp \int_0^\pi \sin(\theta) d\theta \int_0^{2\pi} d\phi \frac{e^{ipr \cos(\theta)}}{2\sqrt{p^2 + m^2}} \rightarrow \text{do } \phi \text{ integral} \\
&= \frac{(2\pi)}{2(2\pi)^3} \int_0^\infty p^2 dp \int_0^\pi \sin(\theta) d\theta \frac{e^{ipr \cos(\theta)}}{\sqrt{p^2 + m^2}} \rightarrow \text{u substitution for } \theta \text{ integral} \\
&= \frac{1}{2(2\pi)^2} \int_0^\infty p^2 dp \int_{-1}^1 du \frac{e^{ipru}}{\sqrt{p^2 + m^2}} \rightarrow \text{do } u \text{ integral} \\
&= \frac{1}{2(2\pi)^2} \int_0^\infty p^2 dp \frac{1}{\sqrt{p^2 + m^2}} \frac{2 \sin(pr)}{pr} \rightarrow \text{simplify} \\
&= \frac{1}{r(2\pi)^2} \int_0^\infty dp \frac{p \sin(pr)}{\sqrt{p^2 + m^2}} \rightarrow \text{use Gradstein and Ryzhik equation 3.754.2} \\
&= \frac{1}{r(2\pi)^2} \left(-\frac{\partial}{\partial r} K_0(rm) \right) \rightarrow \frac{\partial}{\partial r} K_0(rm) = -mK_1(rm) \\
&= \frac{1}{(2\pi)^2} \frac{m}{r} K_1(rm) \rightarrow K_\nu \text{ are the modified Bessel functions.}
\end{aligned}$$

Chapter 3

The Dirac Field

3.1 Lorentz group

(a) The commutation relations of the generators of the group of Lorentz transformations are given by:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i \left(g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} - g^{\nu\sigma} J^{\mu\rho} + g^{\mu\sigma} J^{\nu\rho} \right),$$

where the differential operators $J^{\mu\nu}$ are antisymmetric, i.e., $J^{\nu\mu} = -J^{\mu\nu}$. First, from the definition of L^i , we can calculate the following useful relation:

$$\begin{aligned} L^k &= \frac{1}{2} \epsilon^{klm} J^{lm} \rightarrow \text{multiply by } \epsilon^{ijk} \\ \epsilon^{ijk} L^k &= \frac{1}{2} \epsilon^{ijk} \epsilon^{klm} J^{lm} \\ &= \frac{1}{2} (\delta^{il} \delta^{jm} - \delta^{im} \delta^{jl}) J^{lm} \\ &= \frac{1}{2} (J^{ij} - J^{ji}) \\ &= \frac{1}{2} (J^{ij} + J^{ij}) \\ &= J^{ij}. \end{aligned}$$

The commutation relations between two generators of a rotation are:

$$[L^i, L^j] = \left[\frac{1}{2} \epsilon^{imn} J^{mn}, \frac{1}{2} \epsilon^{jkl} J^{kl} \right]$$

$$\begin{aligned}
&= \frac{1}{4} \epsilon^{imn} \epsilon^{jkl} [J^{mn}, J^{kl}] \\
&= \frac{i}{4} \epsilon^{imn} \epsilon^{jkl} (g^{nk} J^{ml} - g^{mk} J^{nl} - g^{nl} J^{mk} + g^{ml} J^{nk}) \\
&= \frac{i}{4} \epsilon^{imn} (-\epsilon^{jnl} J^{ml} + \epsilon^{jml} J^{nl} + \epsilon^{jkn} J^{mk} - \epsilon^{jkm} J^{nk}) \\
&= \frac{i}{4} \epsilon^{imn} (\epsilon^{jln} J^{ml} + \epsilon^{jml} J^{nl} + \epsilon^{jkn} J^{mk} + \epsilon^{jmk} J^{nk}) \\
&= \frac{i}{4} \left((\delta^{ij} \delta^{ml} - \delta^{il} \delta^{mj}) J^{ml} + (\delta^{ij} \delta^{nl} - \delta^{il} \delta^{nj}) J^{nl} \right. \\
&\quad \left. + (\delta^{ij} \delta^{mk} - \delta^{ik} \delta^{mj}) J^{mk} + (\delta^{ij} \delta^{nk} - \delta^{ik} \delta^{nj}) J^{nk} \right) \\
&= \frac{i}{4} \left(\delta^{ij} J^{ll} - J^{ji} + \delta^{ij} J^{ll} - J^{ji} + \delta^{ij} J^{kk} - J^{ji} + \delta^{ij} J^{kk} - J^{ji} \right) \\
&= \frac{i}{4} \left(0 + J^{ij} + 0 + J^{ij} + 0 + J^{ij} + 0 + J^{ij} \right) \\
&= i J^{ij} \rightarrow \text{use above relation} \\
&= i \epsilon^{ijk} L^k.
\end{aligned}$$

The commutation relations between two generators of a boost are:

$$\begin{aligned}
[K^i, K^j] &= [J^{0i}, J^{0j}] \\
&= i (g^{i0} J^{0j} - g^{00} J^{ij} - g^{ij} J^{00} + g^{0j} J^{i0}) \\
&= i (0 - J^{ij} - 0 + 0) \\
&= -i J^{ij} \rightarrow \text{use above relation} \\
&= -i \epsilon^{ijk} L^k.
\end{aligned}$$

The commutation relations between generators of a rotation and a boost are:

$$\begin{aligned}
[L^i, K^j] &= \left[\frac{1}{2} \epsilon^{ikl} J^{kl}, J^{0j} \right] \\
&= \frac{i}{2} \epsilon^{ikl} (g^{l0} J^{kj} - g^{k0} J^{lj} - g^{lj} J^{k0} + g^{kj} J^{l0}) \\
&= \frac{i}{2} \epsilon^{ikl} (0 - 0 - g^{lj} J^{k0} + g^{kj} J^{l0}) \\
&= \frac{i}{2} (\epsilon^{ikj} J^{k0} - \epsilon^{ijl} J^{l0}) \\
&= \frac{i}{2} (\epsilon^{ikj} (-K^k) - \epsilon^{ijl} (-K^l))
\end{aligned}$$

$$\begin{aligned}
&= \frac{i}{2} \left(\epsilon^{ijk} K^k + \epsilon^{ijl} K^l \right) \rightarrow \text{relabel } l \text{ to } k \text{ in second term} \\
&= i\epsilon^{ijk} K^k.
\end{aligned}$$

The commutation relations between J_+^i and J_-^j are:

$$\begin{aligned}
[J_+^i, J_-^j] &= \left[\frac{1}{2}(L^i + iK^i), \frac{1}{2}(L^j - iK^j) \right] \\
&= \frac{1}{4} \left([L^i, L^j] - i[L^i, K^j] + i[K^i, L^j] + [K^i, K^j] \right) \\
&= \frac{1}{4} \left(i\epsilon^{ijk} L^k - i(i\epsilon^{ijk} K^k) + i(-i\epsilon^{jik} K^k) + (-i\epsilon^{ijk} L^k) \right) \\
&= \frac{1}{4} \left(-i(i\epsilon^{ijk} K^k) + i(i\epsilon^{ijk} K^k) \right) \\
&= 0.
\end{aligned}$$

The commutation relations between $J_\pm^i$ and $J_\pm^j$ are:

$$\begin{aligned}
[J_\pm^i, J_\pm^j] &= \left[\frac{1}{2}(L^i \pm iK^i), \frac{1}{2}(L^j \pm iK^j) \right] \\
&= \frac{1}{4} \left([L^i, L^j] \pm i[L^i, K^j] \pm i[K^i, L^j] - [K^i, K^j] \right) \\
&= \frac{1}{4} \left(i\epsilon^{ijk} L^k \pm i(i\epsilon^{ijk} K^k) \pm i(-i\epsilon^{jik} K^k) - (-i\epsilon^{ijk} L^k) \right) \\
&= \frac{1}{4} \left(i\epsilon^{ijk} L^k \pm i(i\epsilon^{ijk} K^k) \pm i(i\epsilon^{ijk} K^k) + i\epsilon^{ijk} L^k \right) \\
&= \frac{1}{2} \left(i\epsilon^{ijk} L^k \pm i(i\epsilon^{ijk} K^k) \right) \\
&= \frac{1}{2} i\epsilon^{ijk} (L^k \pm iK^k) \\
&= i\epsilon^{ijk} J_\pm^k.
\end{aligned}$$

(b) An infinitesimal Lorentz transformation, under rotations $\boldsymbol{\theta}$ and boosts $\boldsymbol{\beta}$, is given by:

$$\Phi \rightarrow (1 - i\boldsymbol{\theta} \cdot \mathbf{L} - i\boldsymbol{\beta} \cdot \mathbf{K})\Phi.$$

In order to write the transformation laws of the 2-component objects transforming according to the $(j_+, j_-) = (\frac{1}{2}, 0)$ and $(j_+, j_-) = (0, \frac{1}{2})$ representations of the Lorentz group, we first need to determine $\mathbf{L}$ and $\mathbf{K}$ in terms of $\mathbf{J}$. By adding the expressions for $\mathbf{J}_+$ and $\mathbf{J}_-$, we obtain:

$$\mathbf{J}_+ + \mathbf{J}_- = \frac{1}{2}(\mathbf{L} + i\mathbf{K}) + \frac{1}{2}(\mathbf{L} - i\mathbf{K})$$

$$= \mathbf{L}.$$

Similarly, by subtracting the expressions for $\mathbf{J}_+$ and $\mathbf{J}_-$, we obtain:

$$\begin{aligned}\mathbf{J}_+ - \mathbf{J}_- &= \frac{1}{2}(\mathbf{L} + i\mathbf{K}) - \frac{1}{2}(\mathbf{L} - i\mathbf{K}) \\ &= i\mathbf{K}.\end{aligned}$$

In the $(j_+, j_-) = (\frac{1}{2}, 0)$ representation, we have $\mathbf{J}_+$ represented by $\frac{1}{2}\boldsymbol{\sigma}$ and $\mathbf{J}_-$ represented by 0. For the generators of rotations, this gives:

$$\begin{aligned}\mathbf{L} &= \mathbf{J}_+ + \mathbf{J}_- \\ &= \frac{\boldsymbol{\sigma}}{2},\end{aligned}$$

and for the generators of boosts:

$$\begin{aligned}\mathbf{K} &= -i(\mathbf{J}_+ - \mathbf{J}_-) \\ &= -i\frac{\boldsymbol{\sigma}}{2}.\end{aligned}$$

Inserting these expressions for $\mathbf{L}$ and $\mathbf{K}$ into the above expression for an infinitesimal Lorentz transformation, we have:

$$\begin{aligned}\Phi &\rightarrow \left(1 - i\boldsymbol{\theta} \cdot \mathbf{L} - i\boldsymbol{\beta} \cdot \mathbf{K}\right)\Phi \\ &= \left(1 - i\boldsymbol{\theta} \cdot \frac{\boldsymbol{\sigma}}{2} - \boldsymbol{\beta} \cdot \frac{\boldsymbol{\sigma}}{2}\right)\Phi,\end{aligned}$$

which is precisely the transformation ψ_L . In the $(j_+, j_-) = (0, \frac{1}{2})$ representation, we have $\mathbf{J}_+$ represented by 0 and $\mathbf{J}_-$ represented by $\frac{1}{2}\boldsymbol{\sigma}$. For the generators of rotations, this gives:

$$\begin{aligned}\mathbf{L} &= \mathbf{J}_+ + \mathbf{J}_- \\ &= \frac{\boldsymbol{\sigma}}{2},\end{aligned}$$

and for the generators of boosts:

$$\begin{aligned}\mathbf{K} &= -i(\mathbf{J}_+ - \mathbf{J}_-) \\ &= i\frac{\boldsymbol{\sigma}}{2}.\end{aligned}$$

Inserting these expressions for $\mathbf{L}$ and $\mathbf{K}$ into the above expression for an infinitesimal Lorentz transformation, we have:

$$\begin{aligned}\Phi &\rightarrow \left(1 - i\boldsymbol{\theta} \cdot \mathbf{L} - i\boldsymbol{\beta} \cdot \mathbf{K}\right)\Phi \\ &= \left(1 - i\boldsymbol{\theta} \cdot \frac{\boldsymbol{\sigma}}{2} + \boldsymbol{\beta} \cdot \frac{\boldsymbol{\sigma}}{2}\right)\Phi,\end{aligned}$$

which is precisely the transformation ψ_R .

(c) In the $(j_+, j_-) = (\frac{1}{2}, \frac{1}{2})$ representation, the transformation law for the object V that transforms under infinitesimal rotations and boosts is given as:

$$V \rightarrow \left(1 - i\boldsymbol{\theta} \cdot \frac{\boldsymbol{\sigma}}{2} + \boldsymbol{\beta} \cdot \frac{\boldsymbol{\sigma}}{2}\right) V \left(1 + i\boldsymbol{\theta} \cdot \frac{\boldsymbol{\sigma}}{2} + \boldsymbol{\beta} \cdot \frac{\boldsymbol{\sigma}}{2}\right).$$

The object V is a 2×2 matrix, whose parameterization is given as:

$$V = \begin{pmatrix} V^0 + V^3 & V^1 - iV^2 \\ V^1 + iV^2 & V^0 - V^3 \end{pmatrix}.$$

This matrix can be rewritten as a linear combination of the Pauli matrices, that is:

$$\begin{aligned} V &= V^0 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + V^1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + V^2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + V^3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= V^0 \sigma^0 + V^1 \sigma^1 + V^2 \sigma^2 + V^3 \sigma^3 \\ &= V^0 \sigma^0 + V^i \sigma^i \\ &= V^\mu \sigma^\mu, \end{aligned}$$

where $\sigma^0 = \mathbb{I}_{2 \times 2}$, $\boldsymbol{\sigma} = (\sigma^1, \sigma^2, \sigma^3)$, and $\sigma^\mu = (\sigma^0, \boldsymbol{\sigma})$. We can show that V transforms as a 4-vector by comparing the linearized transformation law for V to the expression obtained by direct application of the transformation law for 4-vectors. Expanding the above expression for V , and only keeping the terms linear in $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$, we obtain:

$$\begin{aligned} V &\rightarrow \left(1 - i\boldsymbol{\theta} \cdot \frac{\boldsymbol{\sigma}}{2} + \boldsymbol{\beta} \cdot \frac{\boldsymbol{\sigma}}{2}\right) V \left(1 + i\boldsymbol{\theta} \cdot \frac{\boldsymbol{\sigma}}{2} + \boldsymbol{\beta} \cdot \frac{\boldsymbol{\sigma}}{2}\right) \\ V^\mu \sigma^\mu &\rightarrow \left(1 - \frac{i}{2} \theta^j \sigma^j + \frac{1}{2} \beta^j \sigma^j\right) \left(V^0 \sigma^0 + V^k \sigma^k\right) \left(1 + \frac{i}{2} \theta^l \sigma^l + \frac{1}{2} \beta^l \sigma^l\right) \\ &\approx V^0 \sigma^0 + V^k \sigma^k + \frac{i}{2} V^0 \theta^l \sigma^0 \sigma^l - \frac{i}{2} V^0 \theta^j \sigma^l \sigma^0 + \frac{i}{2} V^k \theta^l \sigma^k \sigma^l - \frac{i}{2} V^k \theta^j \sigma^i \sigma^k \\ &\quad + \frac{1}{2} V^0 \beta^l \sigma^0 \sigma^l + \frac{1}{2} V^0 \beta^j \sigma^j \sigma^0 + \frac{1}{2} V^k \beta^l \sigma^k \sigma^l + \frac{1}{2} V^k \beta^j \sigma^j \sigma^k \\ &= V^\mu \sigma^\mu + \frac{i}{2} V^0 \theta^l [\sigma^0, \sigma^l] + \frac{i}{2} V^k \theta^l [\sigma^k, \sigma^l] + \frac{1}{2} V^0 \beta^l \{\sigma^0, \sigma^l\} + \frac{1}{2} V^k \beta^l \{\sigma^k, \sigma^l\} \\ &= V^\mu \sigma^\mu + 0 + \frac{i}{2} V^k \theta^l (2i\epsilon^{klm} \sigma^m) + \frac{1}{2} V^0 \beta^l (2\sigma^l) + \frac{1}{2} V^k \beta^l (2\delta^{kl} \sigma^0) \\ &= V^\mu \sigma^\mu - \epsilon^{klm} V^k \theta^l \sigma^m + V^0 \beta^l \sigma^l + V^k \beta^k \sigma^0 \rightarrow \text{reorganize and relabel terms} \\ &= V^\mu \sigma^\mu + \beta^k V^k \sigma^0 + \beta^k V^0 \sigma^k - \epsilon^{klm} \theta^l V^k \sigma^m. \end{aligned}$$

The infinitesimal Lorentz transformation law for 4-vectors is parameterized as:

$$V^\alpha \rightarrow \left(\delta^\alpha_\beta - \frac{i}{2} \omega_{\mu\nu} (\mathcal{J}^{\mu\nu})^\alpha_\beta\right) V^\beta,$$

where

$$(\mathcal{J}^{\mu\nu})_{\alpha\beta} = i(\delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha)$$

and $\omega_{\mu\nu}$ is an antisymmetric tensor. Applying this to V , we obtain:

$$\begin{aligned}
V^\mu \sigma^\mu &\rightarrow \left(\delta^\mu{}_\nu + \frac{1}{2} \omega_{\alpha\beta} (g^{\alpha\mu} \delta^\beta{}_\nu - \delta^\alpha{}_\nu g^{\beta\mu}) \right) V^\nu \sigma^\mu \\
&= \delta^\mu{}_\nu V^\nu \sigma^\mu + \frac{1}{2} \omega_{\alpha\beta} g^{\alpha\mu} \delta^\beta{}_\nu V^\nu \sigma^\mu - \frac{1}{2} \omega_{\alpha\beta} \delta^\alpha{}_\nu g^{\beta\mu} V^\nu \sigma^\mu \\
&= V^\mu \sigma^\mu + \frac{1}{2} \omega_{\alpha\nu} g^{\alpha\mu} V^\nu \sigma^\mu - \frac{1}{2} \omega_{\nu\beta} g^{\beta\mu} V^\nu \sigma^\mu \\
&= V^\mu \sigma^\mu + \frac{1}{2} \omega_{\alpha\nu} g^{\alpha\mu} V^\nu \sigma^\mu + \frac{1}{2} \omega_{\alpha\nu} g^{\alpha\mu} V^\nu \sigma^\mu \\
&= V^\mu \sigma^\mu + \omega_{\alpha\nu} g^{\alpha\mu} V^\nu \sigma^\mu \\
&= V^\mu \sigma^\mu + \omega_{0\nu} g^{00} V^\nu \sigma^0 + \omega_{k\nu} g^{k0} V^\nu \sigma^0 + \omega_{0\nu} g^{0k} V^\nu \sigma^k + \omega_{k\nu} g^{kk} V^\nu \sigma^k \\
&= V^\mu \sigma^\mu + \omega_{0\nu} V^\nu \sigma^0 - \omega_{k\nu} V^\nu \sigma^k \\
&= V^\mu \sigma^\mu + \omega_{0k} V^k \sigma^0 - \omega_{k0} V^0 \sigma^k - \omega_{kj} V^j \sigma^k \\
&= V^\mu \sigma^\mu + \omega_{0k} V^k \sigma^0 + \omega_{0k} V^0 \sigma^k - \omega_{kj} V^j \sigma^k \rightarrow \text{let } \beta^k = \omega_{0k} \text{ and } \epsilon^{kjl} \theta^l = \omega_{kj} \\
&= V^\mu \sigma^\mu + \beta^k V^k \sigma^0 + \beta^k V^0 \sigma^k - \epsilon^{kjl} \theta^l V^j \sigma^k \rightarrow \text{relabel dummy indices} \\
&= V^\mu \sigma^\mu + \beta^k V^k \sigma^0 + \beta^k V^0 \sigma^k - \epsilon^{klm} \theta^l V^k \sigma^m,
\end{aligned}$$

which is the same expression as above. Therefore, V transforms as a 4-vector.

3.2 The Gordon identity

In order to derive the Gordon identity, we first determine two necessary relationships from the following properties for gamma matrices:

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}$$

and

$$\sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu] \implies \gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu = -2i\sigma^{\mu\nu}.$$

First,

$$\begin{aligned}
\gamma^\mu \gamma^\nu &= 2g^{\mu\nu} - \gamma^\nu \gamma^\mu \\
&= 2g^{\mu\nu} - \gamma^\mu \gamma^\nu - 2i\sigma^{\mu\nu} \rightarrow \text{move } \gamma^\mu \gamma^\nu \text{ to the LHS and divide through by 2} \\
&= g^{\mu\nu} - i\sigma^{\mu\nu}.
\end{aligned}$$

Similarly,

$$\gamma^\nu \gamma^\mu = 2g^{\mu\nu} - \gamma^\mu \gamma^\nu$$

$$\begin{aligned}
&= 2g^{\mu\nu} - (\gamma^\nu \gamma^\mu - 2i\sigma^{\mu\nu}) \\
&= 2g^{\mu\nu} - \gamma^\nu \gamma^\mu + 2i\sigma^{\mu\nu} \rightarrow \text{move } \gamma^\nu \gamma^\mu \text{ to the LHS and divide through by 2} \\
&= g^{\mu\nu} + i\sigma^{\mu\nu}.
\end{aligned}$$

Now, to derive the Gordon identity, we make explicit use of the Dirac equation and its Hermitian conjugate. The Dirac equation in momentum-space form for plane wave solutions $u(p)$ is given as:

$$0 = (\gamma^\nu p_\nu - m)u(p).$$

The Hermitian conjugate of the Dirac equation for plane wave solutions $u(p')$ is given as:

$$\begin{aligned}
0 &= \left((\gamma^\nu p'_\nu - m)u(p') \right)^\dagger \\
&= u^\dagger(p') \left((\gamma^\nu)^\dagger p'_\nu - m \right) \rightarrow \text{use } (\gamma^\nu)^\dagger = \gamma^0 \gamma^\nu \gamma^0 \\
&= u^\dagger(p') (\gamma^0 \gamma^\nu \gamma^0 p'_\nu - m) \rightarrow \text{multiply from the right by } \gamma^0 \\
&= u^\dagger(p') (\gamma^0 \gamma^\nu \gamma^0 \gamma^0 p'_\nu - m \gamma^0) \rightarrow \text{use } (\gamma^0)^2 = 1 \text{ and } \bar{u}(p') = u^\dagger(p') \gamma^0 \\
&= \bar{u}(p') (\gamma^\nu p'_\nu - m).
\end{aligned}$$

Multiplying the Dirac equation by $\bar{u}(p')\gamma^\mu$ from the left and the Hermitian conjugate Dirac equation by $\gamma^\mu u(p)$ from the right, we obtain:

$$0 = \bar{u}(p')\gamma^\mu \gamma^\nu p_\nu u(p) - m\bar{u}(p')\gamma^\mu u(p)$$

and

$$0 = \bar{u}(p')\gamma^\nu p'_\nu \gamma^\mu u(p) - m\bar{u}(p')\gamma^\mu u(p).$$

Adding these two expressions together, we obtain:

$$\begin{aligned}
2m\bar{u}(p')\gamma^\mu u(p) &= \bar{u}(p')\gamma^\mu \gamma^\nu p_\nu u(p) + \bar{u}(p')\gamma^\nu p'_\nu \gamma^\mu u(p) \\
&= \bar{u}(p') \left[\gamma^\mu \gamma^\nu p_\nu + \gamma^\nu \gamma^\mu p'_\nu \right] u(p) \rightarrow \text{use above relationships} \\
&= \bar{u}(p') \left[(g^{\mu\nu} - i\sigma^{\mu\nu})p_\nu + (g^{\mu\nu} + i\sigma^{\mu\nu})p'_\nu \right] u(p) \\
&= \bar{u}(p') \left[p^\mu - i\sigma^{\mu\nu} p_\nu + p'^\mu + i\sigma^{\mu\nu} p'_\nu \right] u(p) \\
&= \bar{u}(p') \left[(p'^\mu + p^\mu) + i\sigma^{\mu\nu} (p'_\nu - p_\nu) \right] u(p).
\end{aligned}$$

Moving the factor of $2m$ to the right hand side and using $q_\nu = (p'_\nu - p_\nu)$, we obtain the Gordon identity:

$$\bar{u}(p')\gamma^\mu u(p) = \bar{u}(p') \left[\frac{p'^\mu + p^\mu}{2m} + \frac{i\sigma^{\mu\nu} q_\nu}{2m} \right] u(p).$$

3.3 Spinor products

(a) From the problem statement, and working in the Weyl (or chiral) representation of the gamma matrices, u_{L0} is a left-handed spinor for a massless fermion with momentum k_0 . Since u_{L0} must satisfy the Dirac equation, inserting u_{L0} and $m = 0$ gives:

$$\begin{aligned} 0 &= (\gamma^\mu k_{0\mu} - m)u_{L0} \\ &= \gamma^\mu k_{0\mu}u_{L0} \\ &= \not{k}_0 u_{L0}. \end{aligned}$$

Using this result, we can show:

$$\begin{aligned} \not{k}_0 u_{R0} &= \not{k}_0 \not{k}_1 u_{L0} \\ &= \gamma^\mu k_{0\mu} \gamma^\nu k_{1\nu} u_{L0} \\ &= \gamma^\mu \gamma^\nu k_{0\mu} k_{1\nu} u_{L0} \rightarrow \text{use } \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \\ &= (2g^{\mu\nu} - \gamma^\nu \gamma^\mu) k_{0\mu} k_{1\nu} u_{L0} \\ &= 2g^{\mu\nu} k_{0\mu} k_{1\nu} u_{L0} - \gamma^\nu \gamma^\mu k_{0\mu} k_{1\nu} u_{L0} \rightarrow \text{raise index in first term} \\ &= 2k_0^\nu k_{1\nu} u_{L0} - \gamma^\nu k_{1\nu} \gamma^\mu k_{0\mu} u_{L0} \rightarrow k_0^\nu k_{1\nu} = k_0 \cdot k_1 = 0 \text{ by definition} \\ &= -\not{k}_1 \not{k}_0 u_{L0} \rightarrow \not{k}_0 u_{L0} = 0 \text{ from above} \\ &= 0. \end{aligned}$$

For any lightlike p , i.e., $p^2 = 0$, we have:

$$\begin{aligned} \not{p} u_L(p) &= \frac{1}{\sqrt{2p \cdot k_0}} \not{p} \not{p} u_{R0} \\ &= 0. \end{aligned}$$

This follows from the fact that:

$$\begin{aligned} \not{p} \not{p} &= \gamma^\mu p_\mu \gamma^\nu p_\nu \\ &= \gamma^\mu \gamma^\nu p_\mu p_\nu \rightarrow \text{use } \gamma^\mu \gamma^\nu = g^{\mu\nu} - i\sigma^{\mu\nu} \text{ (see Problem 3.2 above for proof)} \\ &= (g^{\mu\nu} - i\sigma^{\mu\nu}) p_\mu p_\nu \rightarrow \text{multiply through by } p_\mu p_\nu \text{ and raise index in first term} \\ &= p^\nu p_\nu - i\sigma^{\mu\nu} p_\mu p_\nu \rightarrow p^\nu p_\nu = p \cdot p = p^2 = 0 \\ &= -i\sigma^{\mu\nu} p_\mu p_\nu \\ &= 0. \end{aligned}$$

This last line follows from the fact that we are contracting indices of a symmetric quantity with an anti-symmetric quantity. When summing over μ and ν , p_μ and p_ν are the components of the 4-vector p , which are ordinary numbers that commute, while the $\sigma^{\mu\nu}$ are matrices that pick up a minus sign when μ and ν are swapped; thus leading to their cancellation. The quantity $\not{p} u_R(p)$ can be shown to be 0 in the same manner for the same reasons.

(b) We are given the 4-vectors $k_0 = (E, 0, 0, -E)$ and $k_1 = (0, 1, 0, 0)$. A left-handed, 4-component spinor for a massless fermion with momentum k_0 can be written in the following form:

$$u_{L0} = \begin{pmatrix} \chi \\ 0 \end{pmatrix},$$

where the two-component object χ is the left-handed Weyl spinor, written as:

$$\chi = \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix},$$

and the lower 2 components of u_{L0} are 0. Starting from the Dirac equation, and working in the Weyl (or chiral) representation of the gamma matrices, we have:

$$\begin{aligned} 0 &= \gamma^\mu k_{0\mu} u_{L0} \\ &= \begin{pmatrix} 0 & \sigma^\mu k_{0\mu} \\ \bar{\sigma}^\mu k_{0\mu} & 0 \end{pmatrix} \begin{pmatrix} \chi \\ 0 \end{pmatrix} \\ &= \bar{\sigma}^\mu k_{0\mu} \chi \rightarrow \text{explicitly write out each term} \\ &= \bar{\sigma}^0 k_{00} \chi + \bar{\sigma}^1 k_{01} \chi + \bar{\sigma}^2 k_{02} \chi + \bar{\sigma}^3 k_{03} \chi \rightarrow \text{insert } \bar{\sigma}^\mu \text{ and } k_{0\mu} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} (E) \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} + \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} (-E) \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \\ &= \begin{cases} E\chi_1 + E\chi_1 \implies 2E\chi_1 = 0 \implies \chi_1 = 0 \\ E\chi_2 - E\chi_2 \implies \chi_2 \text{ is arbitrary.} \end{cases} \end{aligned}$$

Therefore,

$$\chi = \begin{pmatrix} 0 \\ \chi_2 \end{pmatrix}.$$

For this single-particle state, we need to normalize χ so that $\chi^\dagger \chi = 1$. In general, χ_2 may be a complex number, written in polar form as:

$$\chi_2 = |\chi_2| e^{i\theta},$$

where the exponential term is a phase factor. We can take this phase factor to be 1 (i.e., $\theta = 0$) since it is irrelevant for a single-particle state (i.e., the norm remains unchanged and its effects are unobservable). This normalization criteria leaves us with $|\chi_2| = 1$. Therefore,

$$\chi = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Finally, we obtain:

$$u_{L0} = \sqrt{2E} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

where the factor of $\sqrt{2E}$ has been inserted to ensure consistency with the normalization condition:

$$u^\dagger u = 2E.$$

The remaining quantities are then:

$$\begin{aligned}
u_{R0} &= \not{k}_1 u_{L0} \\
&= \gamma^\mu k_{1\mu} u_{L0} \rightarrow \text{explicitly write out each term} \\
&= \gamma^0 k_{10} u_{L0} + \gamma^1 k_{11} u_{L0} + \gamma^2 k_{12} u_{L0} + \gamma^3 k_{13} u_{L0} \rightarrow \text{insert } \gamma^\mu \text{ and } k_{1\mu} \\
&= \begin{pmatrix} 0 & \sigma^1 \\ \bar{\sigma}^1 & 0 \end{pmatrix} u_{L0} \rightarrow \text{insert } \sigma^1 \text{ and } u_{L0} \\
&= \sqrt{2E} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \\
&= \sqrt{2E} \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix}.
\end{aligned}$$

$$\begin{aligned}
u_L(p) &= \frac{1}{\sqrt{2p \cdot k_0}} \not{p} u_{R0} \\
&= \frac{1}{\sqrt{2p \cdot k_0}} \gamma^\mu p_\mu u_{R0} \\
&= \frac{1}{\sqrt{2p \cdot k_0}} \begin{pmatrix} 0 & \sigma^\mu p_\mu \\ \bar{\sigma}^\mu p_\mu & 0 \end{pmatrix} u_{R0} \\
&= \frac{\sqrt{2E}}{\sqrt{2p \cdot k_0}} \begin{pmatrix} 0 & 0 & p_0 + p_3 & p_1 - ip_2 \\ 0 & 0 & p_1 + ip_2 & p_0 - p_3 \\ p_0 - p_3 & -p_1 + ip_2 & 0 & 0 \\ -p_1 - ip_2 & p_0 + p_3 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix} \\
&= \frac{-\sqrt{2E}}{\sqrt{2p \cdot k_0}} \begin{pmatrix} p_0 + p_3 \\ p_1 + ip_2 \\ 0 \\ 0 \end{pmatrix} \rightarrow p \cdot k_0 = p_0 E - (0 + 0 + p_3(-E)) = E(p_0 + p_3) \\
&= \frac{-\sqrt{2E}}{\sqrt{2E(p_0 + p_3)}} \begin{pmatrix} p_0 + p_3 \\ p_1 + ip_2 \\ 0 \\ 0 \end{pmatrix}
\end{aligned}$$

$$= \frac{-1}{\sqrt{(p_0 + p_3)}} \begin{pmatrix} p_0 + p_3 \\ p_1 + ip_2 \\ 0 \\ 0 \end{pmatrix}.$$

$$\begin{aligned} u_R(p) &= \frac{1}{\sqrt{2p \cdot k_0}} \not{p} u_{L0} \\ &= \frac{1}{\sqrt{2p \cdot k_0}} \gamma^\mu p_\mu u_{L0} \\ &= \frac{1}{\sqrt{2p \cdot k_0}} \begin{pmatrix} 0 & \sigma^\mu p_\mu \\ \bar{\sigma}^\mu p_\mu & 0 \end{pmatrix} u_{L0} \\ &= \frac{\sqrt{2E}}{\sqrt{2p \cdot k_0}} \begin{pmatrix} 0 & 0 & p_0 + p_3 & p_1 - ip_2 \\ 0 & 0 & p_1 + ip_2 & p_0 - p_3 \\ p_0 - p_3 & -p_1 + ip_2 & 0 & 0 \\ -p_1 - ip_2 & p_0 + p_3 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \\ &= \frac{\sqrt{2E}}{\sqrt{2p \cdot k_0}} \begin{pmatrix} 0 \\ 0 \\ -p_1 + ip_2 \\ p_0 + p_3 \end{pmatrix} \rightarrow p \cdot k_0 = p_0 E - (0 + 0 + p_3(-E)) = E(p_0 + p_3) \\ &= \frac{\sqrt{2E}}{\sqrt{2E(p_0 + p_3)}} \begin{pmatrix} 0 \\ 0 \\ -p_1 + ip_2 \\ p_0 + p_3 \end{pmatrix} \\ &= \frac{1}{\sqrt{(p_0 + p_3)}} \begin{pmatrix} 0 \\ 0 \\ -p_1 + ip_2 \\ p_0 + p_3 \end{pmatrix}. \end{aligned}$$

(c) We can explicitly compute the spinor products as follows:

$$\begin{aligned} s(p_1, p_2) &= \bar{u}_R(p_1) u_L(p_2) \\ &= u_R^\dagger(p_1) \gamma^0 u_L(p_2) \\ &= \frac{1}{\sqrt{(p_{10} + p_{13})}} \frac{-1}{\sqrt{(p_{20} + p_{23})}} \\ &\quad \times \begin{pmatrix} 0 \\ 0 \\ -p_{11} - ip_{12} \\ p_{10} + p_{13} \end{pmatrix}^T \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} p_{20} + p_{23} \\ p_{21} + ip_{22} \\ 0 \\ 0 \end{pmatrix} \end{aligned}$$

$$= \frac{(p_{11} + ip_{12})(p_{20} + p_{23}) - (p_{10} + p_{13})(p_{21} + ip_{22})}{\sqrt{(p_{10} + p_{13})(p_{20} + p_{23})}}.$$

$$\begin{aligned} t(p_1, p_2) &= \bar{u}_L(p_1)u_R(p_2) \\ &= u_L^\dagger(p_1)\gamma^0 u_R(p_2) \\ &= \frac{-1}{\sqrt{(p_{10} + p_{13})}} \frac{1}{\sqrt{(p_{20} + p_{23})}} \\ &\quad \times \begin{pmatrix} p_{10} + p_{13} \\ p_{11} - ip_{12} \\ 0 \\ 0 \end{pmatrix}^T \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ -p_{21} + ip_{22} \\ p_{20} + p_{23} \end{pmatrix} \\ &= \frac{(p_{10} + p_{13})(p_{21} - ip_{22}) - (p_{11} - ip_{12})(p_{20} + p_{23})}{\sqrt{(p_{10} + p_{13})(p_{20} + p_{23})}}. \end{aligned}$$

Additionally, we can show that:

$$\begin{aligned} (s(p_2, p_1))^* &= \left(\frac{(p_{21} + ip_{22})(p_{10} + p_{13}) - (p_{20} + p_{23})(p_{11} + ip_{12})}{\sqrt{(p_{20} + p_{23})(p_{10} + p_{13})}} \right)^* \\ &= \frac{(p_{21} - ip_{22})(p_{10} + p_{13}) - (p_{20} + p_{23})(p_{11} - ip_{12})}{\sqrt{(p_{20} + p_{23})(p_{10} + p_{13})}} \\ &= t(p_1, p_2). \end{aligned}$$

$$\begin{aligned} -s(p_2, p_1) &= \frac{-(p_{21} + ip_{22})(p_{10} + p_{13}) + (p_{20} + p_{23})(p_{11} + ip_{12})}{\sqrt{(p_{20} + p_{23})(p_{10} + p_{13})}} \\ &= \frac{(p_{20} + p_{23})(p_{11} + ip_{12}) - (p_{21} + ip_{22})(p_{10} + p_{13})}{\sqrt{(p_{20} + p_{23})(p_{10} + p_{13})}} \\ &= s(p_1, p_2). \end{aligned}$$

$$\begin{aligned} |s(p_1, p_2)|^2 &= s(p_1, p_2)^* s(p_1, p_2) \\ &= \frac{(p_{11} - ip_{12})(p_{20} + p_{23}) - (p_{10} + p_{13})(p_{21} - ip_{22})}{\sqrt{(p_{10} + p_{13})(p_{20} + p_{23})}} \\ &\quad \times \frac{(p_{11} + ip_{12})(p_{20} + p_{23}) - (p_{10} + p_{13})(p_{21} + ip_{22})}{\sqrt{(p_{10} + p_{13})(p_{20} + p_{23})}} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{(p_{10} + p_{13})(p_{20} + p_{23})} \\
&\quad \times \left((p_{11} - ip_{12})(p_{20} + p_{23})(p_{11} + ip_{12})(p_{20} + p_{23}) \right. \\
&\quad - (p_{11} - ip_{12})(p_{20} + p_{23})(p_{10} + p_{13})(p_{21} + ip_{22}) \\
&\quad - (p_{10} + p_{13})(p_{21} - ip_{22})(p_{11} + ip_{12})(p_{20} + p_{23}) \\
&\quad \left. + (p_{10} + p_{13})(p_{21} - ip_{22})(p_{10} + p_{13})(p_{21} + ip_{22}) \right) \\
&= \frac{(p_{11} - ip_{12})(p_{11} + ip_{12})(p_{20} + p_{23})}{(p_{10} + p_{13})} + \frac{(p_{10} + p_{13})(p_{21} - ip_{22})(p_{21} + ip_{22})}{(p_{20} + p_{23})} \\
&\quad - (p_{11} - ip_{12})(p_{21} + ip_{22}) - (p_{21} - ip_{22})(p_{11} + ip_{12}) \\
&= \frac{(p_{11}^2 + p_{12}^2)(p_{20} + p_{23})}{(p_{10} + p_{13})} + \frac{(p_{10} + p_{13})(p_{21}^2 + p_{22}^2)}{(p_{20} + p_{23})} \\
&\quad - 2p_{11}p_{21} - 2p_{12}p_{22} \rightarrow \text{use } p^2 = p_0^2 - p_1^2 - p_2^2 - p_3^2 = 0 \\
&= \frac{(p_{10} + p_{13})(p_{10} - p_{13})(p_{20} + p_{23})}{(p_{10} + p_{13})} + \frac{(p_{10} + p_{13})(p_{20} + p_{23})(p_{20} - p_{23})}{(p_{20} + p_{23})} \\
&\quad - 2p_{11}p_{21} - 2p_{12}p_{22} \\
&= (p_{10} - p_{13})(p_{20} + p_{23}) + (p_{10} + p_{13})(p_{20} - p_{23}) - 2p_{11}p_{21} - 2p_{12}p_{22} \\
&= p_{10}p_{20} + p_{10}p_{23} - p_{13}p_{20} - p_{13}p_{23} + p_{10}p_{20} - p_{10}p_{23} + p_{13}p_{20} - p_{13}p_{23} \\
&\quad - 2p_{11}p_{21} - 2p_{12}p_{22} \\
&= 2p_{10}p_{20} - 2p_{11}p_{21} - 2p_{12}p_{22} - 2p_{13}p_{23} \\
&= 2(p_{10}p_{20} - p_{11}p_{21} - p_{12}p_{22} - p_{13}p_{23}) \\
&= 2p_1 \cdot p_2.
\end{aligned}$$

3.4 Majorana fermions

(a)

(b)

(c)

(d)

(e)

3.5 Supersymmetry

(a)

(b)

(c)

3.6 Fierz transformations

(a)

(b)

(c)

3.7 Discrete symmetries P , C , and T

(a)

(b)

(c)

3.8 Bound states

(a)

(b)

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(a)

(b)

(c)

(d)

(e)

(f)

4.2 Decay of a scalar particle

Something...

4.3 Linear sigma model

(a)

(b)

(c)

(d)

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(a)

(b)

(c)

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(a)

(b)

(c)

5.4 Positronium lifetimes

(a)

(b)

(c)

(d)

5.5 Physics of a massive vector boson

(a)

(b)

(c)

(d)

5.6 Extending spinor products to external photons

(a)

(b)

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(a)

(b)

(c)

(d)

(e)

6.3 Exotic contributions to $g - 2$

(a)

(b)

(c)

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(a)

(b)

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(a)

(b)

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(a)

(b)

(c)

(d)

(e)

(f)

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